

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

Table of Contents

Time-Domain analysis

Time-Domain vs Spectral analysis: main tools

The Correlation Function

The Auto-Correlation Function

The partial ACF (PACF)

Discrete Correlation Function

Z-transformed DCF

Ergodic processes

The Structure Function

Different stochastic processes can be categorized based on their ACF/PSD

Stationarity

Types of Stationarity

Covariance matrix

White Noise

Random walk (with or without drift)

Moving average (of the first order)

Autoregressive (of the first order)

General linear processes

Backshift operator

Difference operator

MA(q) and AR(p)

ARMA(p,q) processes

ARIMA(p,d,q)

State space models

Reference & Material

Further Material

Credits

Course Flow

TIME-DOMAIN ASTROPHYSICS

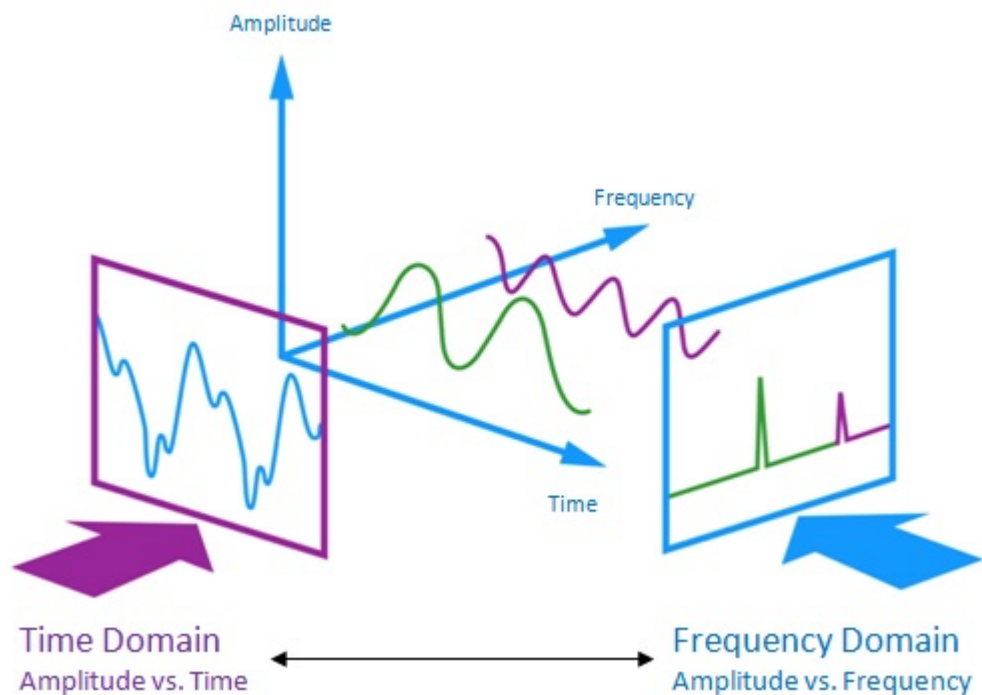
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Time-Domain analysis

- One can derive inferences about the spectral content of a time series by a spectral analysis.
- An analogous (i.e. complementary) analysis can be carried out in the time domain.
- Traditionally, analyses in the temporal domain are often aimed at deriving predictions, while in the spectral domain one often looks for periodicity, etc.
- There are indeed no fundamental technical reasons for that.



Time-Domain vs Spectral analysis: main tools

	Time Domain	Frequency Domain
	x_t linear combination of past	x_t linear combination of periodic components
Object of interest	population ACF	Spectral density
Data analysis tool	sample ACF	Periodogram
		Identify dominant frequency(ies)

The Correlation Function

- One of the main statistical tools for the analysis of stochastic variability is the autocorrelation function. It represents a specialized case of the correlation function of two functions, $f(t)$ and $g(t)$, scaled by their standard deviations, and defined at time lag as:

$$CF(\Delta t) = \frac{\lim_{T \rightarrow \infty} \int_{(T)} f(t)g(t + \Delta t)dt}{\sigma_f \sigma_g}$$

- where σ_f and σ_g are standard deviations of $f(t)$ and $g(t)$, respectively.
- With this normalization, the correlation function is unity for $\Delta t = 0$, without normalization by standard deviation, the above expression is equal to the covariance function.

The Auto-Correlation Function

- It is assumed that both f and g are statistically weakly stationary functions (more on this topic later).
- With $f(t) = g(t) = y(t)$, the autocorrelation of $y(t)$ defined at time lag is:

$$ACF(\Delta t) = \frac{\lim_{T \rightarrow \infty} \int_{(T)} y(t)y(t + \Delta t)dt}{\sigma_y^2}$$

- The autocorrelation function and the PSD of function $y(t)$ are Fourier pairs; this fact is known as the Wiener–Khinchin theorem and applies to stationary random processes.
- The sample auto-correlation function is defined as:

$$ACF(k) = \frac{\sum_{t=1}^{n-k} (X_t - \bar{X})(X_{t+k} - \bar{X})}{\sum_{t=1}^k (X_t - \bar{X})^2}$$

- where the numerator is just the sample auto-covariance function and the denominator the sample variance.
- ACF(k) is often called a correlogram.
- The ACF is fundamental measure of the serial correlation in a time series.
- It is defined for evenly spaced time-series, yet there are generalization to irregularly sampled data.
- Under the null hypothesis that the time series has no correlated structure and the population ACF is zero for all lags except for ACF(0) which is always unity.
- The distribution of the sample is asymptotically normal with mean $-1/n$ and variance $1/n$, i.e. the distribution of the null case ACF is $ACF(k) = \mathcal{N}(-1/n, 1/n)$.
- However, this holds under the assumption one is testing whether a time-series is white noise, i.e. the residuals of a fit. Else, a different formula, based on the assumption that a time-series can be described as a moving average process (see later), is known as Bartlett's formula:

$$\sigma_{ACF(k)} \approx (1/\sqrt{N}) \sqrt{1 + 2 \sum_{i=1}^{k-1} ACF(i)^2}.$$

- Essentially, Bartlett's formula refines the confidence interval calculation for ACF by accounting for the potential dependence between ACF values in non-white noise processes, particularly those following a moving average structure.
- Simple quantities like the sample mean are valid estimates of the population mean for stationary processes, but its uncertainty is not the standard value when autocorrelation is present:

$$\widehat{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \left[1 + 2 \sum_{k=1}^{n-1} (1 - k/n) ACF(k) \right]$$

- Qualitatively, this can be understood as due to the decrease of independent measurements.
- This is indeed one very important result, although often (wrongly) ignored.
- Knowing the ACF one can compute the so-called integrated auto-correlation time $\tau_{X,int}$:

$$\tau_{X,int} = \int_0^\infty ACF(\tau) d\tau$$

- This allows one to define the *effective* number of samples in our dataset $N_{\text{eff}} = (N/2)\tau_{X,int}$ so that the variance can be written as $\widehat{Var}(\bar{X}_n) = \sigma^2/N_{\text{eff}}$.
- The ACF is *non-negative definite*, i.e.: $\sum_{i=1}^n \sum_{j=1}^n \alpha_j ACF(i-j) \alpha_j \geq 0$ for all positive integers n and vectors $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)' \in \mathbb{R}^n$. In fact,

$$\sum_{i=1}^n \sum_{j=1}^n \alpha_j ACF(i-j) \alpha_j = \text{Var} \left(\sum_{i=1}^n \alpha_j x_j \right).$$

The partial ACF (PACF)

- The PACF at lag k gives the autocorrelation at value k removing the effects of correlations at shorter lags.
- The value of the p -th coefficient $PACF(p)$ is found by successively fitting autoregressive models (more on this topic later) with order $1, 2, \dots, p$ and setting the last coefficient of each model to the PACF parameter (this again will be clearer later).
- For instance, for a stationary AR(2) process:

$$PACF(2) = \frac{ACF(2) - ACF(1)^2}{1 - ACF(1)^2}$$

- The partial autocorrelation function of a stationary time series can be calculated by using the Durbin–Levinson Algorithm:

$$\phi_{n,n} = \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}$$

- where $\phi_{n,k} = \phi_{n-1,k} - \phi_{n,n} \phi_{n-1,n-k}$ for $1 \leq k \leq n-1$ and $\rho(n)$ is the ACF.

Discrete Correlation Function

- The discrete correlation function is a procedure for computing the autocorrelation function that avoids interpolating an unevenly spaced dataset onto a regular grid.
- The method can treat both autocorrelation within one time series or cross-correlation between two unevenly spaced time series.
- Consider two datasets (x_i, t_{x_i}) and (z_j, t_{z_j}) with $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$ points, respectively. For autocorrelation within a single dataset, let $z = x$.
- Construct two matrices, the **unbinned discrete correlation function** (UDCF) and its associated time lags:

$$UDCF_{i,j} = \frac{(x_i - \bar{x})(z_j - \bar{z})}{\sigma_x \sigma_z} \quad \Delta_{ij} = t_j - t_i$$

- The UDCF is then grouped into a univariate function of lag-time τ by collecting the $M(\tau)$ data pairs with lags falling within the interval $\tau - \Delta\tau/2 \leq \Delta t_{ij} < \tau + \Delta\tau/2$.
- The resulting discrete correlation function (DCF) and its variance are:

$$DCF(\tau) = \frac{1}{M(\tau)} \sum_{k=1}^{M(\tau)} UDCF_{ij} \quad \text{Var}(\tau) = \frac{1}{(M(\tau) - 1)^2} \sum_{k=1}^{M(\tau)} [UDCF_{ij} - DCF(\tau)]^2$$

Z-transformed DCF

- The drawback of the DCF is that its sample distribution of is known to be very skewed and far from normal.

- If bins have an equal number of points (i.e. bins can have different length), The bin distribution becomes approximately binomial and, if a Fisher z-transform is applied:

$$z(t) = \frac{1}{2} \ln \left(\frac{1 + \text{DCF}(\tau)}{1 - \text{DCF}(\tau)} \right)$$

- The $z(\tau)$ values are now normally distributed with known mean and variance (see Alexander 1997).

Tip

This is a very intriguing topic. I just mention it in passing.

Ergodic processes

- We know, or learn soon, that the autocovariance function plays a key role in the construction of a model for our time series.
 - Thus, an important question is: Is it possible to obtain a 'good' estimate of the autocovariance function?
 - The question is not trivial, since we observe only one realization of the process.
 - In general, with a single realization we are unable to estimate a moment function of a stochastic process.
 - Try to think to the simple mean. A possible way to estimate the mean would be to average the each single realization of the random process generating your data.
 - But, in case one only realization is available, we can only average the obtained time-series, but we could wonder whether this estimate of the average of the parent distribution is *good*.
 - The answer is yes, if the stationary process is **ergodic**.
-
- A stationary stochastic process is said ergodic, with respect to a given population moment, if the sample (or time) moment for a single finite realization of length T converges in quadratic mean to the population moment as T increases to ∞ .
 - Ergodicity must be related directly to the particular population moment, e.g. mean value, covariance, etc.

While this is a fascinating topic in statistics, it is clear it is also beyond our purpose here. Yet, it is important to note that not all stationary processes are ergodic.

- For instance, consider a stationary process $\{x_t; t \in \mathbb{Z}\}$, with $x_t = A \forall t \in \mathbb{Z}$, where A is a random variable with mean 3 and variance 7.
- The process is not mean-ergodic since the sample mean $\hat{x}_T$ does not converge to the parental mean as $T \rightarrow \infty$.
- The ergodicity is a matter of information contained in a single realization of a long duration of the process.

The Structure Function

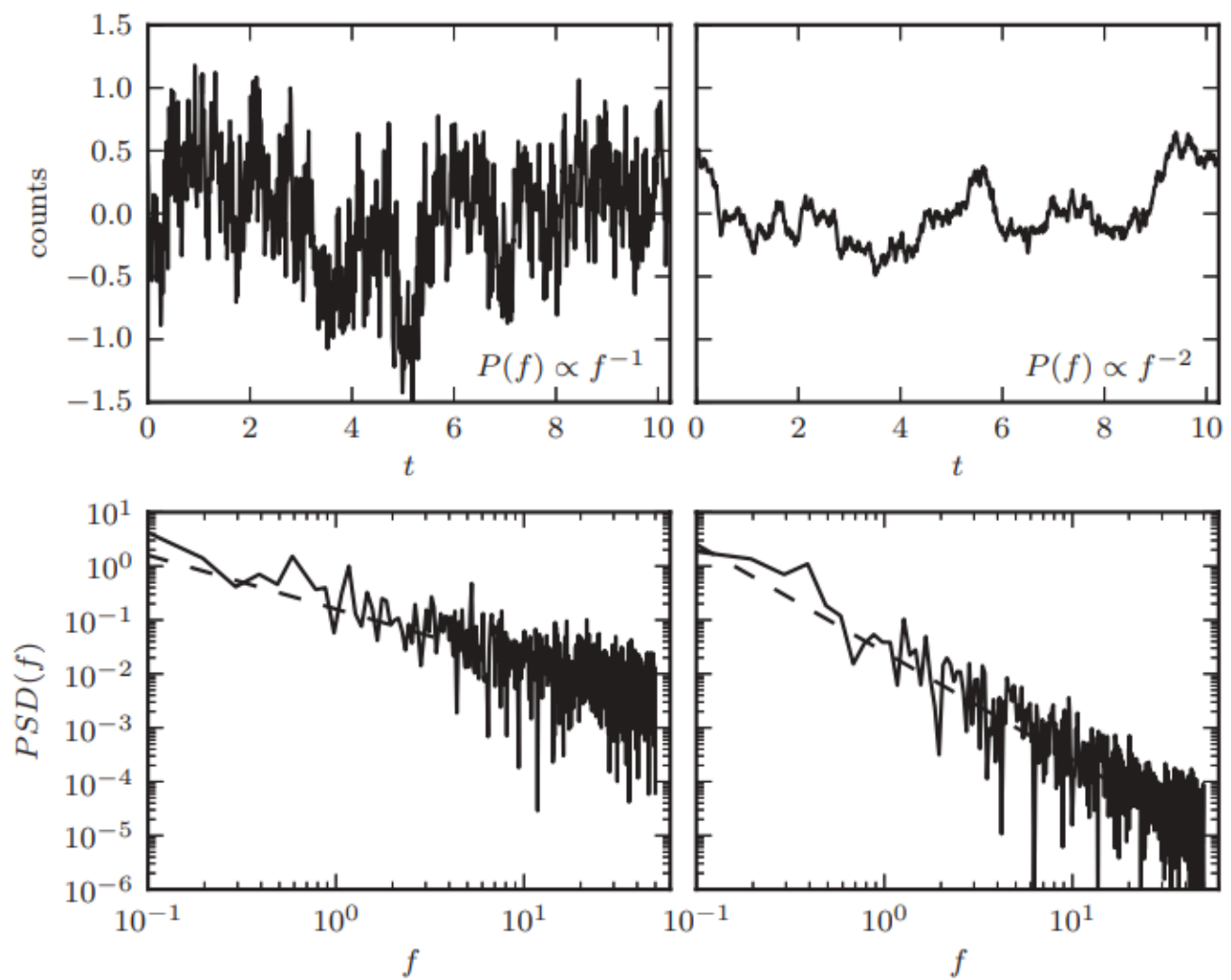
- The structure function (sometimes called the Kolmogorov structure function) is again a measure of autocorrelation.
- The q-th order structure function is:

$$D^q(\tau) = \langle |x(t) - x(t + \tau)|^q \rangle$$

- where the angular brackets $\langle \rangle$ indicate an average over the time series and q is the order of the structure function (not necessarily integer).
- The q = 2 structure function is also called the variogram.
- When a dataset has a characteristic time-scale of variation τ_c , the structure function is small at τ shorter than τ_c , rises rapidly to a high level around τ_c , and stays at this plateau for longer τ .
- If a structure function exhibits a power-law dependence on lag, $D_q(\tau) \propto \tau^\alpha$, the time series has a multi-fractal behavior. In this case, the Wiener–Khinchin theorem links the power-law dependence of the q = 2 structure function to the power-law dependence of the Fourier power spectrum.
- The structure function with q = 2 can also be seen as:

$$SF(\Delta t) = SF_\infty [1 - ACF(\Delta t)]^{1/2}$$

- where SF_∞ is the standard deviation of the time series evaluated over an infinitely large time interval (or at least much longer than any characteristic timescale τ).
 - The structure function with q=2 is equal to the standard deviation of the distribution of the difference of $y(t_2) - y(t_1)$ evaluated at many different t_1 and t_2 .
- As mentioned before, when the structure function $SF \propto t^\alpha$, then the $PSD \propto 1/f^{1+2\alpha}$.
- The slope of the PSD directly affects the shape of the related time-series:



Different stochastic processes can be categorized based on their ACF/PSD

- A stochastic process with $1/f^2$ spectrum is known as random walk (if discrete) or Brownian motion (or, more accurately, Wiener process) if continuous. These physically occur when the value being observed is subjected to a series of independent changes of similar size. It's also sometimes called as "red noise". Quasar variability (for instance) exhibits $1/f^2$ properties at high frequencies (that is, short time scales, below a year or so).
- A stochastic process with $1/f$ spectrum are sometimes called "long-term memory processes" (also sometimes know as "pink noise"). They have equal energy at all octaves (or over any other logarithmic frequency interval). This type of process has infinite variance and an undefined mean (similar to a Lorentzian distribution).
- A process with a constant PSD is frequently referred to as "white noise", i.e., it has equal intensity at all frequencies. This is a process with no memory, each measurement is independent of all others. White noise is identically and independently distributed (I.I.D.).

Exercise: cross-correlation of data from the Covid19 outbreak in Italy

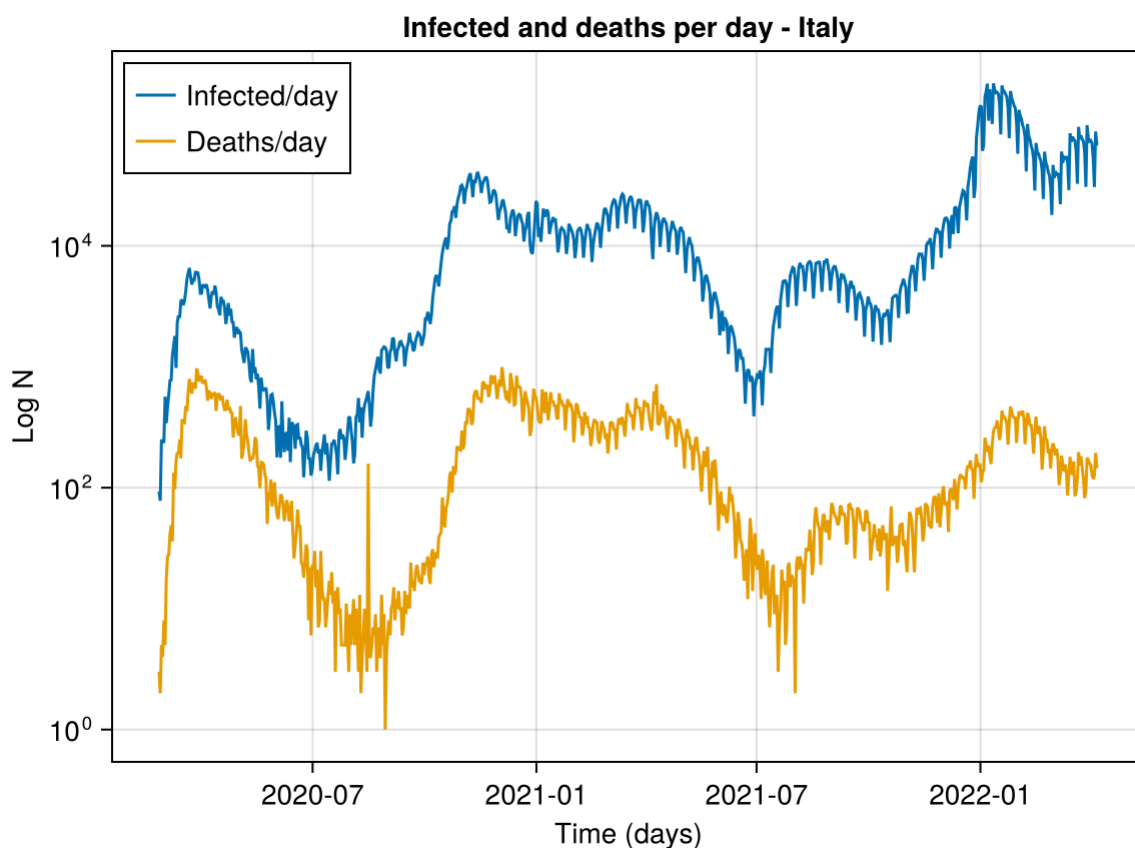
- Italian data downloaded from: <https://github.com/pcm-dpc/COVID-19> and updated till April 07, 2022.

	Date	Hospitalised	In ICU	All Hospitalised	Home Isolation	Infected	Delta Infected	New Infecto
1	2020-02-24T18:00:00	101	26	127	94	221	0	221
2	2020-02-25T18:00:00	114	35	150	162	311	90	93
3	2020-02-26T18:00:00	128	36	164	221	385	74	78
4	2020-02-27T18:00:00	248	56	304	284	588	203	250
5	2020-02-28T18:00:00	345	64	409	412	821	233	238

- We need to slightly rearrange the data in order to compute the number of deaths per day.

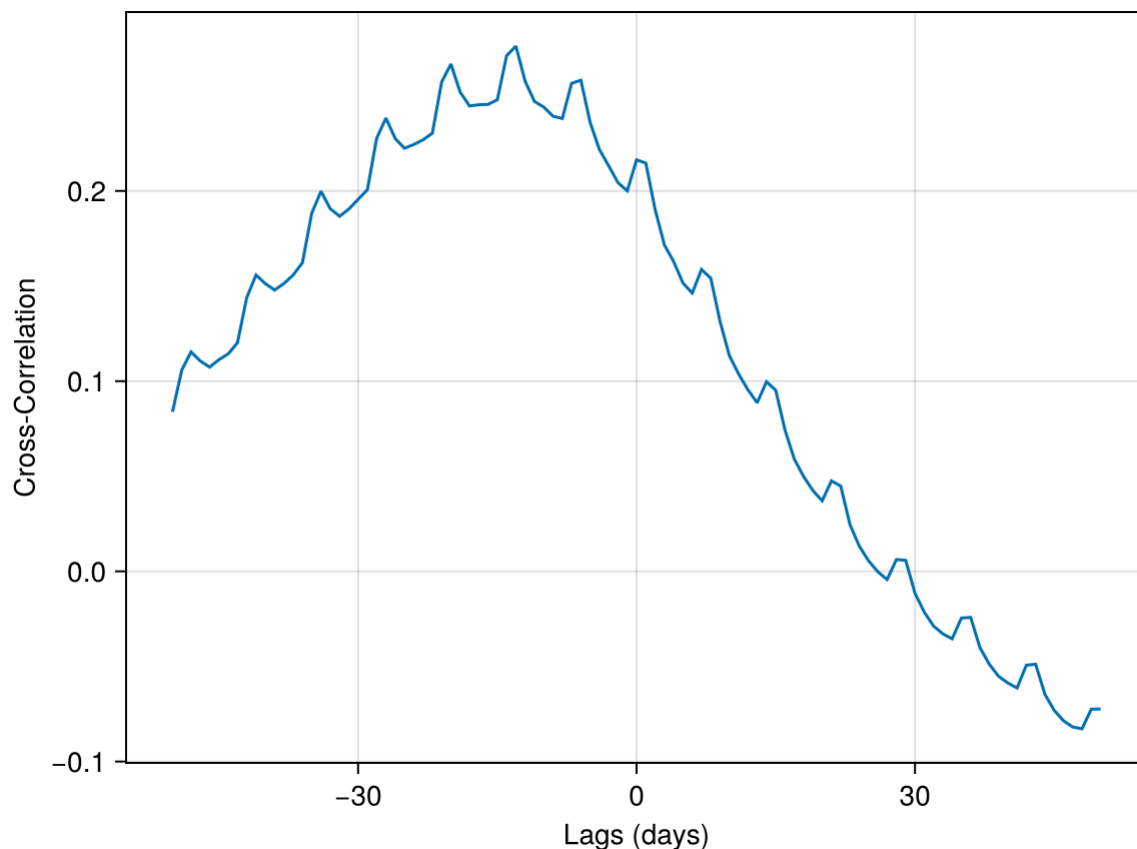
```

1 begin
2   dDead = diff(nep[!,"Dead"])
3   nep[!,"dDead"] = pushfirst!(dDead,0)
4 end;
```



- There is a clear correlation between the number of infected people and deaths per day. Superposed to a seasonal, and shorter, variability together with "spikes" likely due to irregularities in recording the data.
- Lets compute a cross-correlation between these two datasets in order to try to quantify a delay.

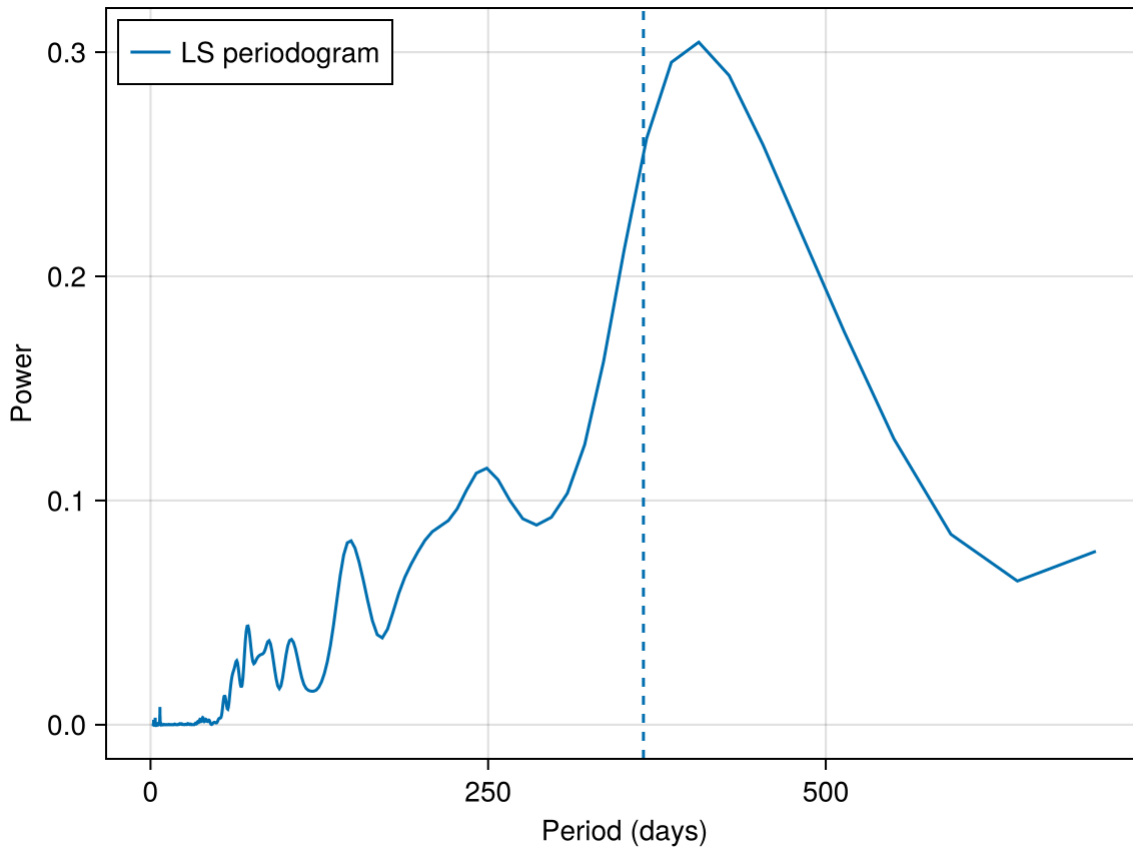
Disclaimer. This is just an exercise. Please, do not underestimate the complexity of a proper epidemiological study.



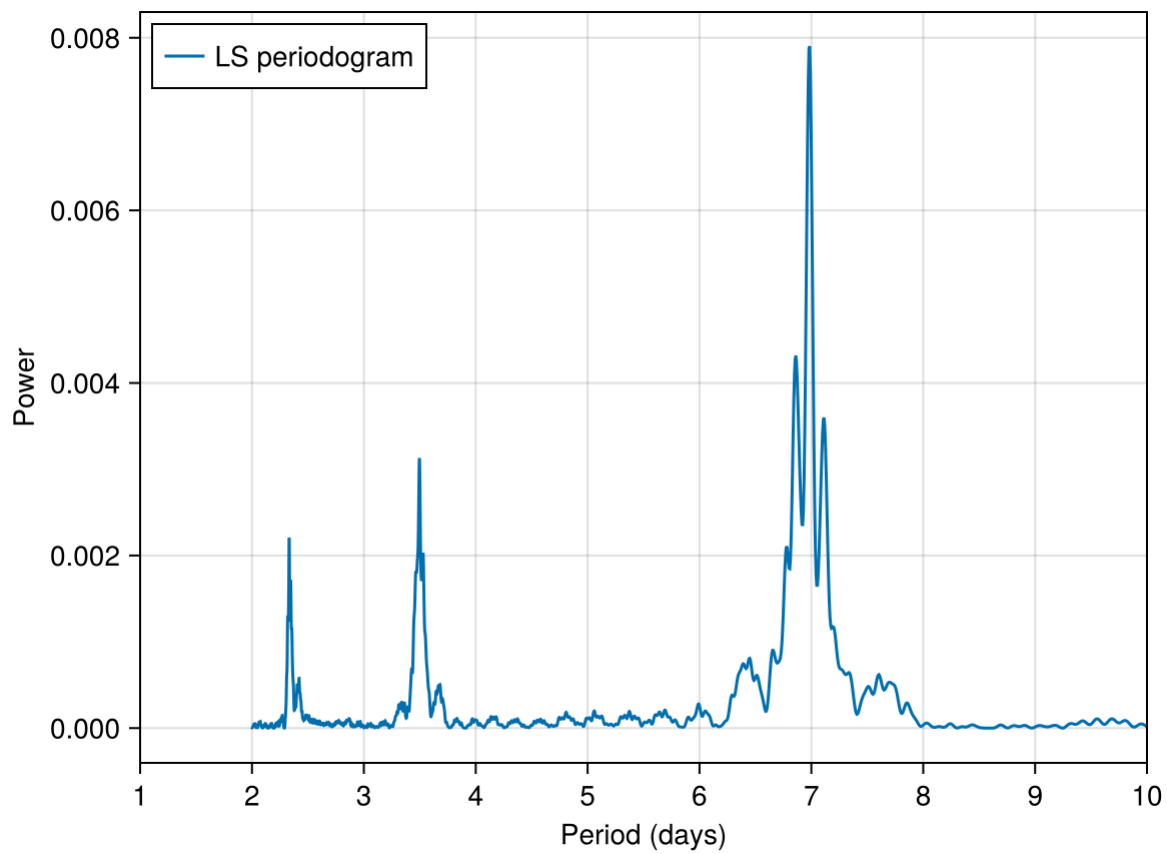
- A ~13 day delay appears, that is quite reasonable.
- Of course, a much more meaningful analysis would have required to separate the infected people basing upon age, sex, etc.

Epidemic periodicity?

- The former plot might suggest some sort of quasi-periodicity of the outbreak.
- Let's see whether a Lomb-Scargle analysis produce interesting results:



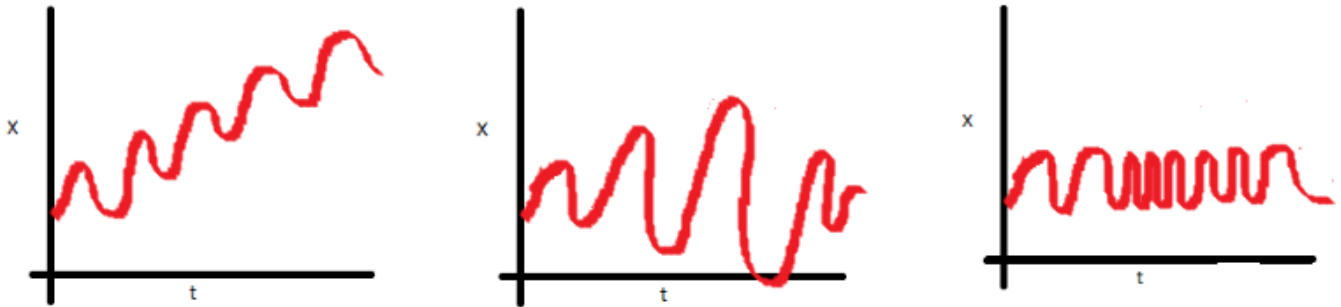
- The LS periodogram shows a considerable power at long periods, with a large peak at about 400 days. This is likely the summer/winter cycle that did not reproduce exactly.
- The time-series shows features at shorter periods too.



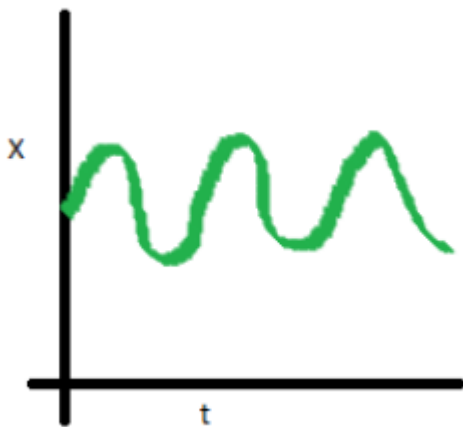
- A distinct peak at a week is clearly visible, indicating the "periodicity" in the collection of data!

Stationarity

- This is a fundamental concept. A stationary time-series is a dataset where it is “meaningful” to draw predictions.
- A stationary series is one in which the properties – mean, variance and covariance, do not vary with time.
 - Let us understand this using an intuitive example. Consider the three plots shown below:



- In the first plot, we can clearly see that the mean varies (increases) with time which results in an upward trend. Thus, this is a non-stationary series. For a series to be classified as stationary, it should not exhibit a trend.
- Moving on to the second plot, there certainly isn't a trend in the series, but the variance of the series is a function of time. As mentioned previously, a stationary series must have a constant variance.
- The the third plot. The spread becomes closer as the time increases, which implies that the covariance is a function of time.
- The three examples shown above represent non-stationary time series.



- In this case, the mean, variance and covariance are constant with time. This is what a stationary time series looks like.
- Most statistical models require the series to be stationary to make effective and precise predictions.

Tip

Important definition!

- Let's now try to define “stationarity” in a more formal way.

- A stochastic process $\mathbf{X}_t; t = 0, \pm 1, \dots$ is stationary if a joint distribution of $\mathbf{X}_t, \dots, \mathbf{X}_{t+k}$ is same as a joint distribution of $\mathbf{X}_0, \dots, \mathbf{X}_k$ for all t and all k.
 - This is typically called strict stationarity.
- A stochastic process $\mathbf{X}_t; t = 0, \pm 1, \dots$ is weakly (or second order) stationary if, for all t:

$$\mathbb{E}(\mathbf{X}_t) = \mu$$

- and $\text{Cov}(\mathbf{X}_t, \mathbf{X}_{t+h}) = \text{Cov}(\mathbf{X}_0, \mathbf{X}_h) = \mathbf{C}_X(h)$ is a function of h only, i.e. it does not depend on t.
 - $\mathbf{C}_X$ is also known as autocovariance of X. $\mathbf{C}_X(h)/\mathbf{C}_x(0)$ is known as autocorrelation.
- Let's recall that: $\text{Cov}(\mathbf{X}, \mathbf{Y}) \equiv \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])]$
- It can be shown that:
 - If $\mathbf{X}_t$ has finite variance, strictly stationary implies $\mathbf{X}_t$ is weakly stationary.
 - If $\mathbf{X}_t$ is second order stationary and Gaussian it implies $\mathbf{X}_t$ is also strictly stationary.
- $\mathbf{X}_t$ is Gaussian if for each $t_1, \dots, t_k$ the vector $(\mathbf{X}_{t_1}, \dots, \mathbf{X}_{t_k})^T$ has a Multivariate Normal Distribution.
- The process $\mathbf{X}_t$ is said to have a stationary covariance if:

$$\text{Cov}(\mathbf{X}_t, \mathbf{X}_s) = \text{Cov}(\mathbf{X}_{t+1}, \mathbf{X}_{s+1}) = \text{Cov}(\mathbf{X}_{t+2}, \mathbf{X}_{s+2}) \dots, \text{etc.}$$

Exercise about time-series stationarity tests

- First read and pre-process a dataset reporting the number of passengers on airplanes in the USA for several years.
 - This is a simple yet interesting dataset for this kind of exercise.

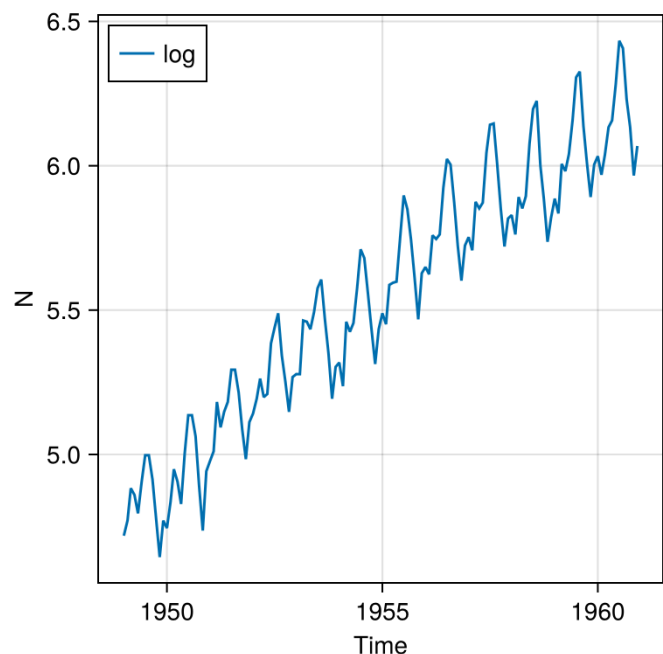
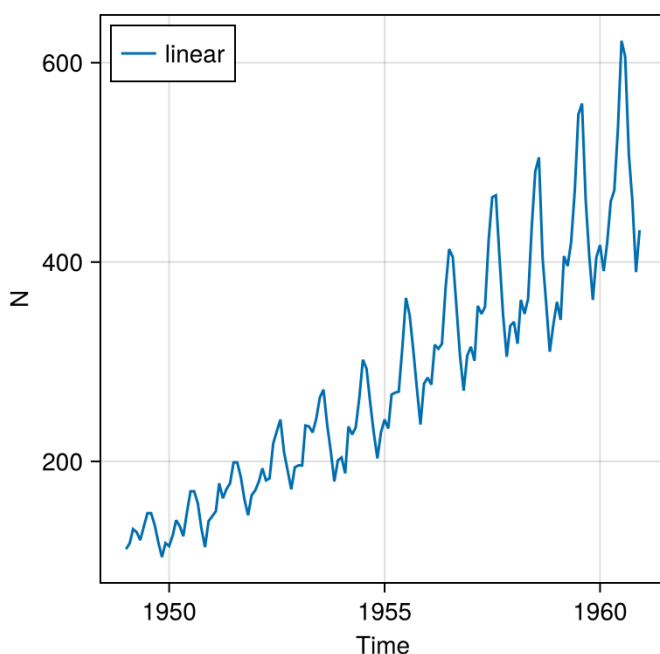
	Month	Passengers
1	1949-01-01	112
2	1949-02-01	118
3	1949-03-01	132
4	1949-04-01	129
5	1949-05-01	121

```

1 begin
2   url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-
   passengers.csv"
3
4   train = CSV.File(Downloads.download(url)) |> DataFrame
5   first(train,5)
6 end

```

- We want to determine whether a given series is stationary or not and deal with it accordingly.
- Let's first try with a simple, yet always useful, visual test.



- We see that there a clear trend, i.e. the mean is varying.
- It is anyway useulf to look for more formal tests.

Statistical tests

- We can use statistical tests like the unit root stationary tests.
 - Unit root indicates that the statistical properties of a given series are not constant with time, which is the condition for stationary time series.
- Suppose we have a time series:

$$y_t = a * y_{t-1} + \varepsilon_t$$

- where y_t is the value at the time instant t and ε_t is the error term. In order to calculate y_t we need the value of y_{t-1} , which is:

$$y_{t-1} = a * y_{t-2} + \varepsilon_{t-1}$$

- If we do that for all observations, the value of y_t will come out to be:

$$y_t = a^n * y_{t-n} + \sum \varepsilon_{t-i} * a^i$$

- If the value of a is 1 (unit) in the above equation, then the predictions will be equal to the y_{t-n} and sum of all errors from $t-n$ to t , which means that the variance will increase with time.
- This is known as unit root in a time series. The unit root tests check the presence of unit root in the series by checking if value of $a=1$.
- The ADF (Augmented Dickey-Fuller) is one of the most commonly used unit root stationary tests:
 - It can be used to determine the presence of unit root in the series, and hence help us understand if the series is stationary or not. The null and alternate hypothesis of this test are:
 - Null Hypothesis: The series has a unit root (value of $a=1$). Alternate Hypothesis: The series has no unit root.
 - If we fail to reject the null hypothesis, we can say that the series is non-stationary.

Augmented Dickey-Fuller unit root test

Population details:

```
parameter of interest: coefficient on lagged non-differenced variable
value under h_0:      0
point estimate:       -0.00316343
```

Test summary:

```
outcome with 95% confidence: fail to reject h_0
p-value:                     0.5556
```

Details:

```
sample size in regression:      142
number of lags:                 1
ADF statistic:                  -0.352401
Critical values at 1%, 5%, and 10%: adjoint([-2.58144, -1.94302, -1.6151])
```

```
1 ADFTest(train[!,"Passengers"],:none,1)
```

- The ADF tests gives the following results – test statistic, p value and the critical value at 1%, 5% , and 10% confidence intervals.
- If the test statistic is less than the critical value, we can reject the null hypothesis and the series can be stationary. When the test statistic is greater than the critical value, we fail to reject the null hypothesis, which means the series is not stationary.

- In our above example, the test statistic $\gg$ critical value, which implies that the series is not stationary, as expected.

Types of Stationarity

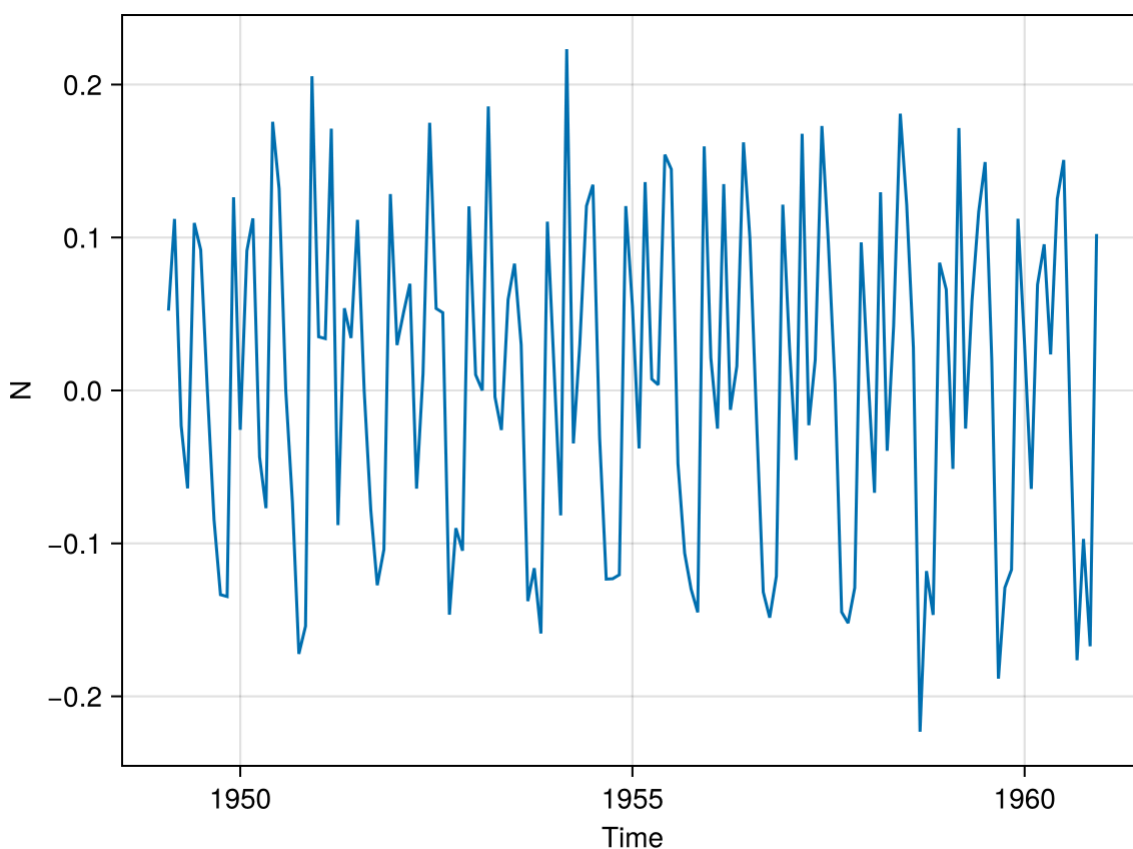
- **Strict Stationary:** A strict stationary series satisfies the mathematical definition of a stationary process. For a strict stationary series, the mean, variance and covariance are not the function of time. The aim is to convert a non-stationary series into a strict stationary series for making predictions.
- **Trend Stationary:** A series that has no unit root but exhibits a trend is referred to as a trend stationary series. Once the trend is removed, the resulting series will be strict stationary.
- **Difference Stationary:** A time series that can be made strict stationary by differencing falls under difference stationary. ADF test is also known as a difference stationarity test.

Exercise about making a time-series stationary

- Is it possible to "stationarize" a time-series?
 - In several cases, it is. For instance, if the non-stationarity is due to a trend, etc.
- Trends can be removed by fitting a smooth version of a time-series or, often, by differenciation, i.e. we compute the difference of consecutive terms in the series. Differencing is typically performed to get rid of the varying mean: $y'_t = y_t - y_{(t-1)}$, where y_t is the value at a time t.
- Let's apply a differentiation on our series and plotting the results:

Lag for difference:

```
1 begin
2   train[!,"Passengers_log"] = log.(train[!,"Passengers"])
3   passengereasonal = train[!,"Passengers_log"] -
   ShiftedArrays.lag(train[!,"Passengers_log"],airlag)
4 end;
```



Augmented Dickey-Fuller unit root test

Population details:

parameter of interest:	coefficient on lagged non-differenced variable
value under h_0 :	0
point estimate:	-0.927795

Test summary:

outcome with 95% confidence:	reject h_0
p-value:	<1e-14

Details:

sample size in regression:	141
number of lags:	1
ADF statistic:	-8.81571
Critical values at 1%, 5%, and 10%:	adjoint([-2.58155, -1.94303, -1.61509])

- With just lag 1 the trend is removed. However, it is possible to compute the difference with longer lags. This is often called *seasonal differencing*.
 - In seasonal differencing, instead of calculating the difference between consecutive values, we calculate the difference between an observation and a previous observation from the same "season": $y_t' = y_t - y_{(t-n)}$.
-
- A lag of 12 months shows us the residuals after that the trend is removed. The resulting curve is not really stationary.
 - We will go back to this time-series later.

Covariance matrix

- Let's take the opportunity to formally (re-)define the covariance matrix Σ , whose components are $\sigma_{i,j} = \text{Cov}(X_i, X_j)$:

$$\Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}$$

- This kind of matrix is also a “Toeplitz” matrix (it has useful properties for massive computations).

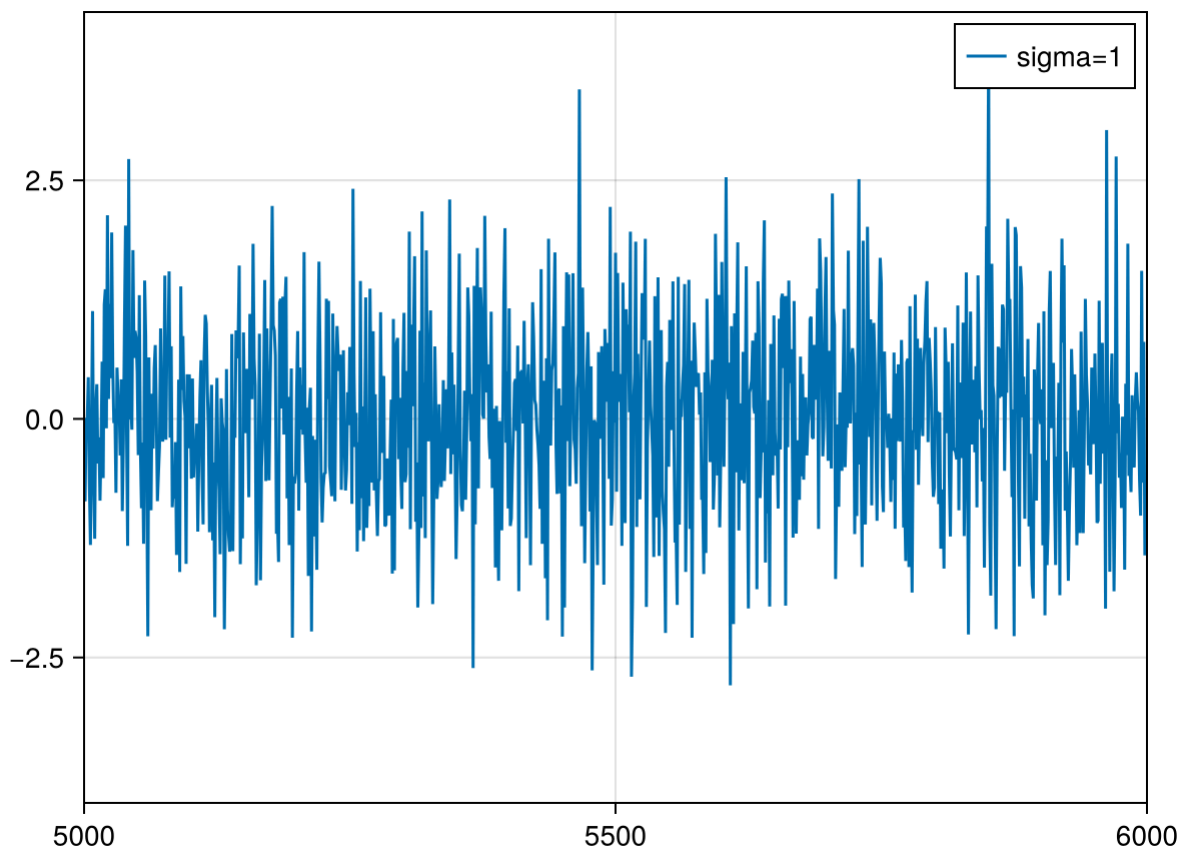
Tip

Let's now study some simple linear process of interest.

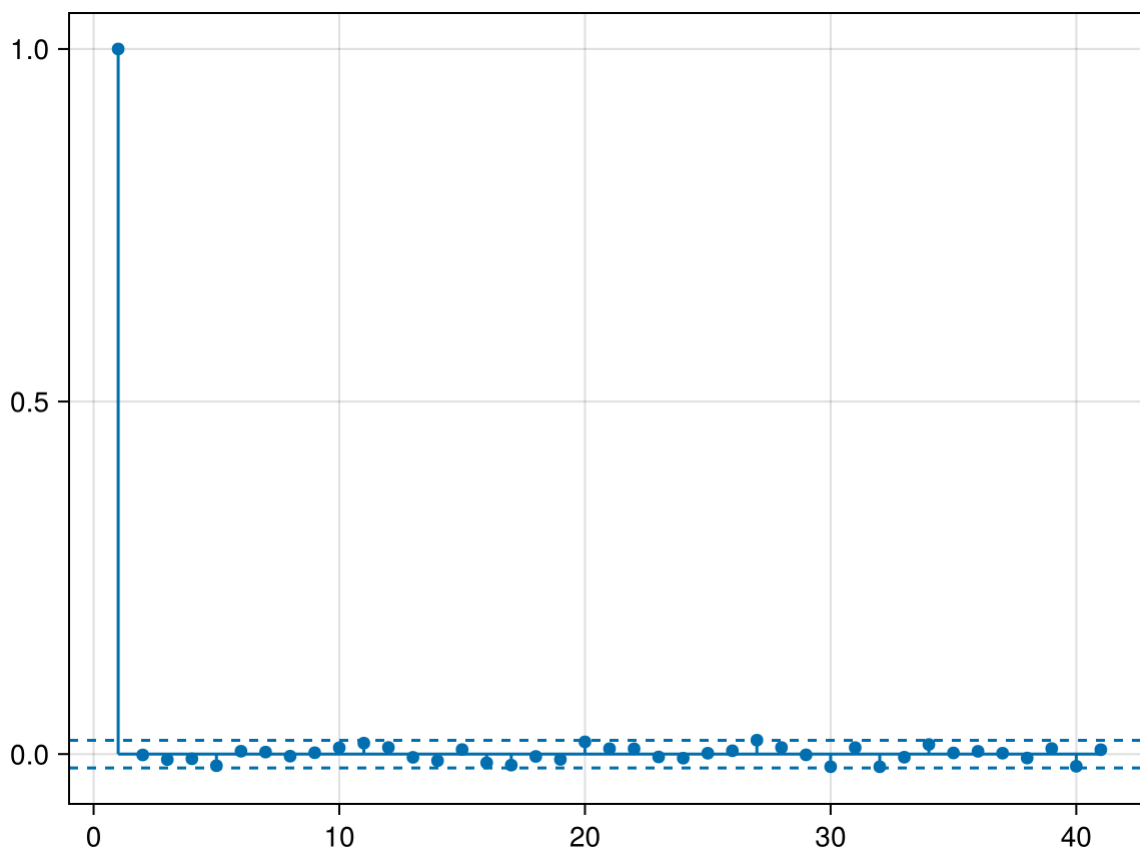
White Noise

- $\{w_t\}$ is a white noise process if w_t are uncorrelated and identically distributed random variables with:
 - $\mathbb{E}[w_t] = 0$ and $\text{Var}[w_t] = \sigma^2$, for all t .
- If $\{w_t\}$ are Normally (Gaussian) distributed, we call this Gaussian white noise.
- A white noise process is stationary.
- This is an example with $\sigma^2 = 1$:

```
1 begin
2   N9 = 10000
3
4   d9 = Normal()
5   x9 = rand(d9, N9)
6 end;
```



- The ACF of a white noise shows a distinct and easy to recognize pattern: there should be no significant correlation for any lag but 0.



Random walk (with or without drift)

- A random walk, with drift, can be defined as:

$$x_t = \delta + x_{t-1} + w_t$$

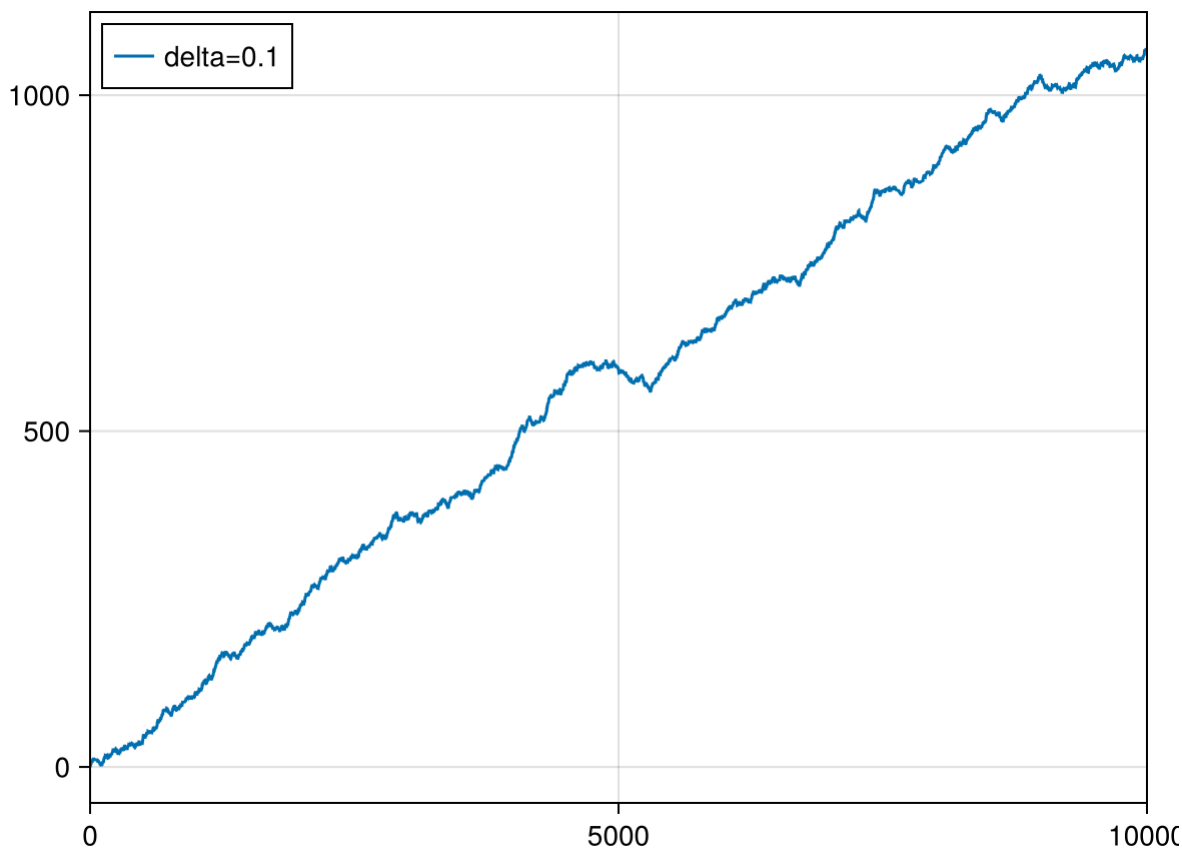
- where δ is a constant, w_t is a white noise process, and we assume for simplicity $x_0 = 0$.
- The equation can be rewritten as: $x_t = t\delta + \sum_{j=1}^t w_j$ simply writing $x_1 = \delta + w_1$, $x_2 = \delta + x_1 + w_2 = \delta + \delta + w_1 + w_2$, etc.
- Is a random walk (with or without drift) stationary?
- The expectation value is simple to compute: $\mathbb{E}[x_t] = \mathbb{E}[t\delta + \sum_{j=1}^t w_j] = t\delta$, since a sum of random noise should give a zero result.
 - Therefore, unless $\delta = 0$ the expectation value depends on t .
- For the covariance we need some more steps:
 - $\text{Cov}(x_t, x_{t+h}) = \text{Cov}[t\delta + \sum_{j=1}^t w_j, (t+h)\delta + \sum_{j=1}^{t+h} w_j]$.
 - Recalling that $\text{Cov}(\sum_i x_i, \sum_j x_j) = \sum_i \sum_j \text{Cov}(x_i, x_j)$ and that $\text{Cov}(X, X) = \text{Var}(X)$ we get that:
 - $\text{Cov}(x_t, x_{t+h}) = \text{Cov}[t\delta, (t+h)\delta] + \text{Cov}[\sum_{j=1}^t w_j, (t+h)\delta] + \text{Cov}[t\delta, \sum_{j=1}^{t+h} w_j] + \text{Cov}[\sum_{j=1}^t w_j, \sum_{j=1}^{t+h} w_j]$.
 - All terms but the last are 0 since they include constant terms. We finally have: $\text{Cov}(x_t, x_{t+h}) = t\text{Var}(w_t) = t\sigma^2$.
 - Clearly non-stationary (for any δ).
- And a random walk with drift ($\delta = 0.1$) can be:

```

1 begin
2     N10 = 10000
3     if driftvl
4         delta10 = 0.1
5     else
6         delta10 = 0.
7     end
8
9     d10 = Normal()
10    sigma10 = rand(d10, N10)
11
12    x10 = zeros(N10)
13    for i in range(2,N10)
14        x10[i] = x10[i-1]+sigma10[i]+delta10
15    end
16 end

```

With drift? ☒



Moving average (of the first order)

- A moving average process of the first order, or MA(1), is defined as:

$$x_t = \beta_1 w_{t-1} + w_t$$

- where again w_t is a white noise process.

- The expectation value of a MA(1) process is:

$$\mathbb{E}[x_t] = \mathbb{E}[\beta_1 w_{t-1} + w_t] = \beta_1 \mathbb{E}[w_{t-1}] + \mathbb{E}[w_t] = 0.$$

- The variance is: $\text{Var}[x_t] = \text{Var}[\beta_1 w_{t-1} + w_t] = \beta_1^2 \text{Var}[w_{t-1}] + \text{Var}[w_t] = \sigma^2(1 + \beta_1^2)$.

- The covariance is: $\text{Cov}[x_t, x_{t+h}] = \text{Cov}[\beta_1 w_{t-1} + w_t, \beta_1 w_{t+h-1} + w_{t+h}]$.

- Now, if $h = 1$ we have: $\text{Cov}[x_t, x_{t+1}] = \text{Cov}[\beta_1 w_{t-1} + w_t, \beta_1 w_t + w_{t+1}] = \text{Cov}[\beta_1 w_{t-1}, \beta_1 w_t] + \text{Cov}[\beta_1 w_{t-1}, w_{t+1}] + \text{Cov}[w_t, \beta_1 w_t] + \text{Cov}[w_t, w_{t+1}]$. All terms with different index are zero (no covariance between independent white noise processes), and therefore: $\text{Cov}[x_t, x_{t+1}] = \beta_1^2 \sigma^2$.

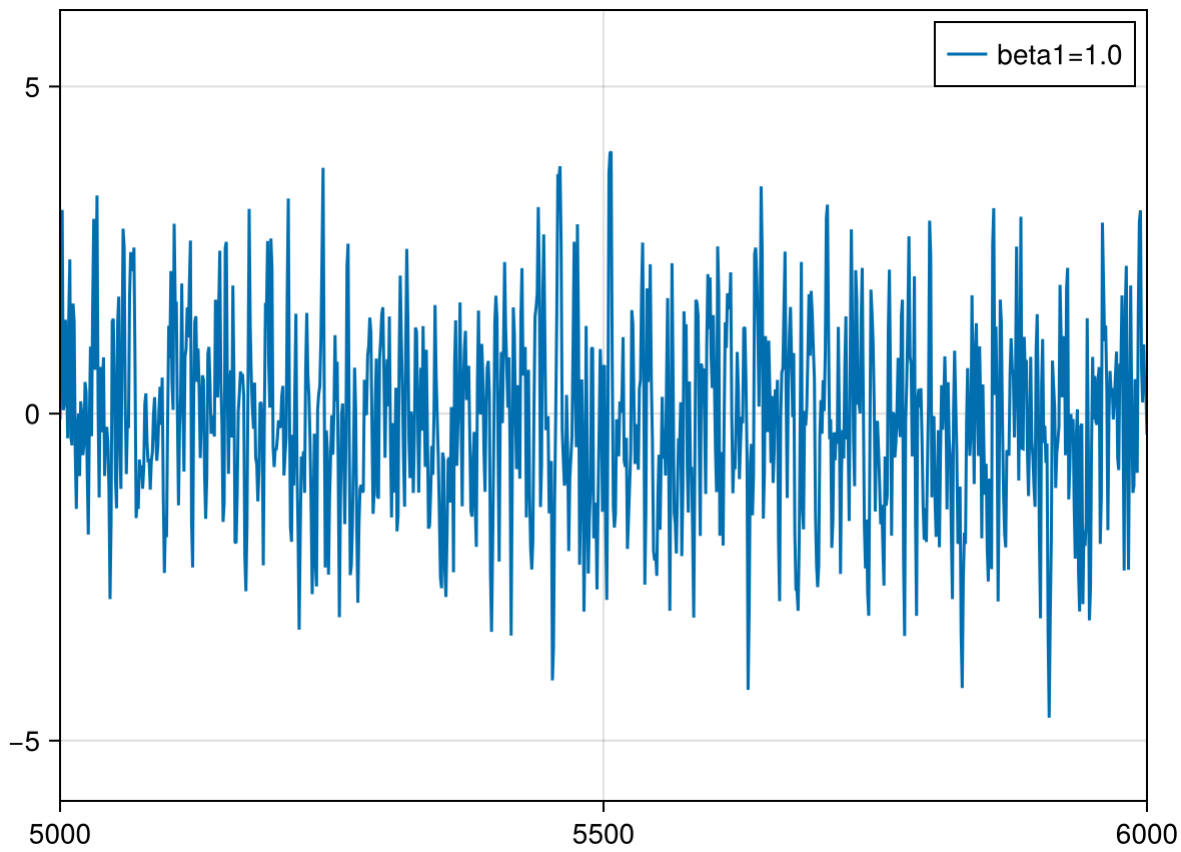
- If $h > 1$ all terms are zero and therefore: $\text{Cov}[x_t, x_{t+1}] = 0$.

- A MA(1) with $\beta_1 = 1$ can be:

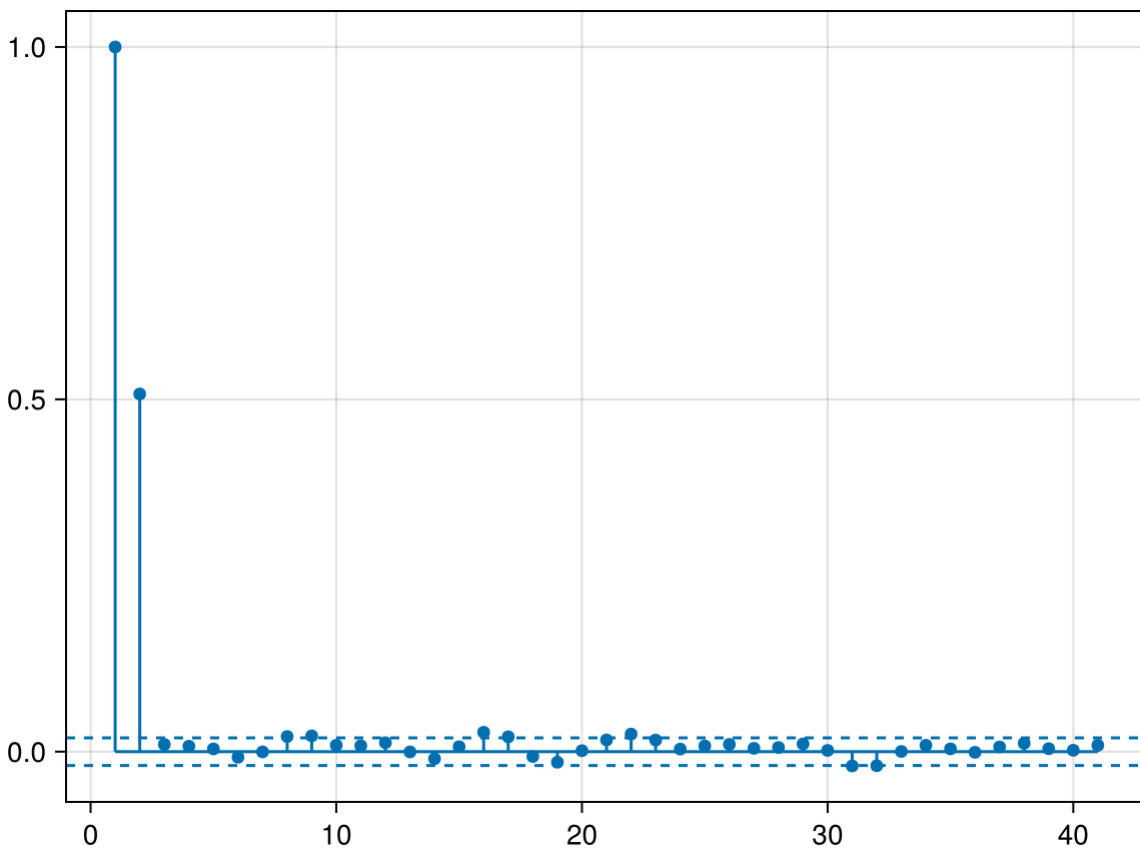
```

1 begin
2     N12 = 10000
3     beta112 = 1.
4
5     d12 = Normal()
6     sigma12 = rand(d12, N12)
7
8     x12 = zeros(N12)
9     for i in range(2,N12)
10         x12[i] = sigma12[i]+beta112*sigma12[i-1]
11     end
12 end

```

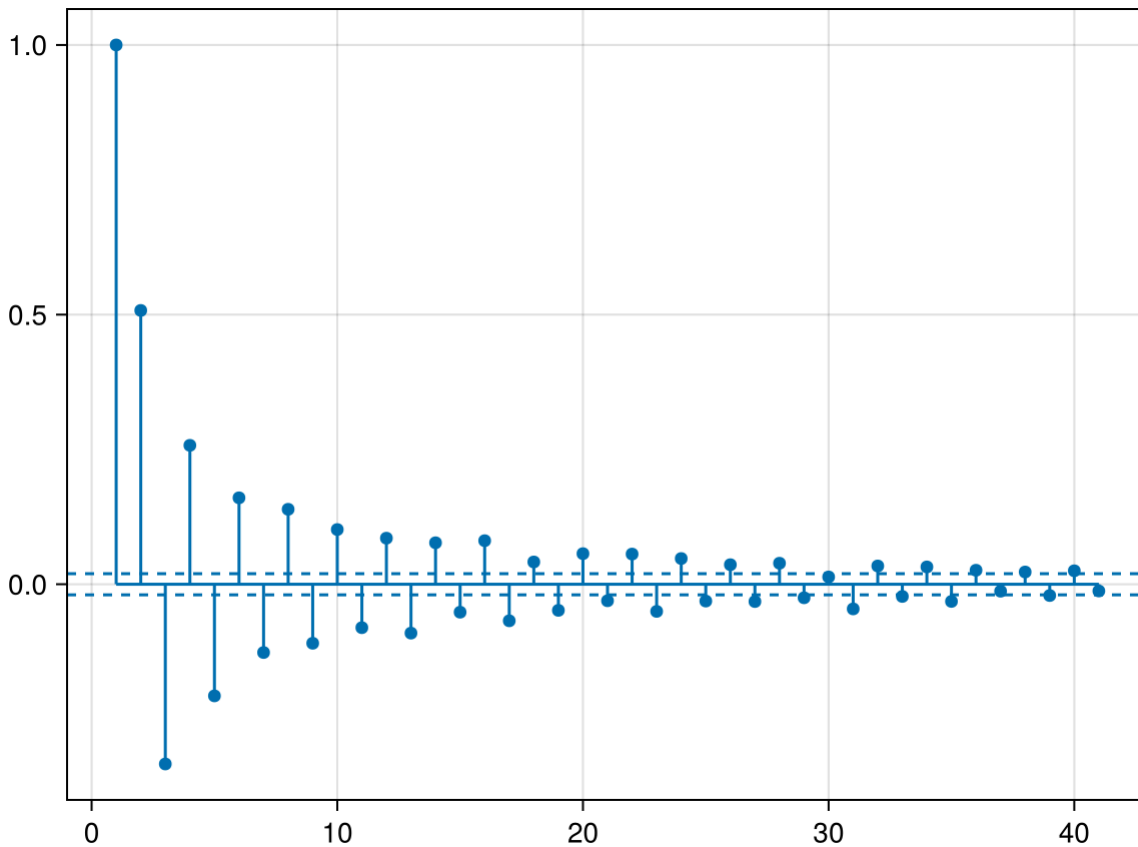


- Since the process is a sum of independent noise processes it is visually very close to a random noise.
- Let's analyse the ACF of the previous process and discover this is not fully the case:



- Apart from lag=0, the only lag showing correlation significantly different from 0 is lag=1, the order of the process. This is not, as we will see, by chance.

- And now also the PACF:



- That is, instead, showing a regularly decreasing correlation increasing the lag.

Autoregressive (of the first order)

- An autoregressive process of the first order, or AR(1), is defined as:

$$x_t = \alpha_1 x_{t-1} + w_t$$

- and again w_t is a white noise process.

- The process can be rewritten as follows:

$$x_t = \alpha_1 x_{t-1} + w_t = \alpha_1 (\alpha_1 x_{t-2} + w_{t-1}) + w_t = \alpha_1^t x_0 + \sum_{i=0}^{t-1} \alpha_1^i w_{t-i}$$

- The expectation value is: $\mathbb{E}[x_t] = \mathbb{E}[\alpha_1^t x_0 + \sum_{i=0}^{t-1} \alpha_1^i w_{t-i}] = \alpha_1^t \mathbb{E}[x_0]$. This is zero only if the starting position is zero.

- The variance is: $\text{Var}[x_t] = \text{Var}[\alpha_1 x_{t-1} + w_t] = \alpha_1^2 \text{Var}[x_{t-1}] + \text{Var}[w_t] \Rightarrow \text{Var}[x_t] = \frac{\sigma^2}{1-\alpha_1^2}$. This relation gives a valid variance only if $|\alpha_1| < 1$.

- The covariance is:

$$\text{Cov}[x_t, x_{t+h}] = \text{Cov}[x_t, \alpha_1^{t+h} x_0 + \sum_{i=0}^{t+h-1} \alpha_1^i w_{t+h-i}] = \text{Cov}[x_t, \alpha_1^h x_t + \sum_{i=0}^{h-1} \alpha_1^i w_{t+h-i}] =$$

, again valid if $|\alpha_1| < 1$.

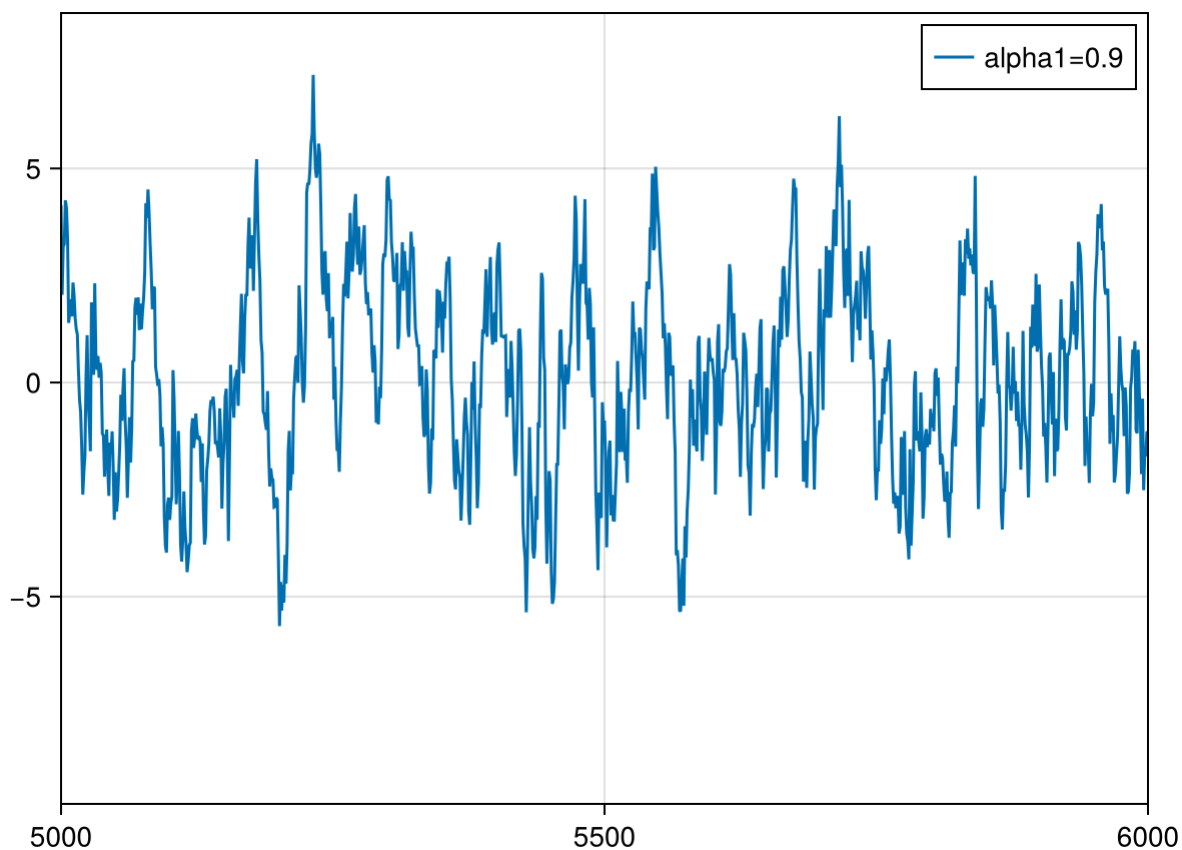
- And, therefore, the correlation turns out to be: $\text{Corr}[x_t, x_{t+h}] = \frac{\text{Cov}[x_t, x_{t+h}]}{\text{Std}[x_t] \text{Std}[x_{t+h}]} = \alpha_1^h$.
- In AR(1) process, the value of α_1 determines whether the AR(1) process is stationary.

- Let's see a AR(1) with $\alpha_1 = 0.9$.

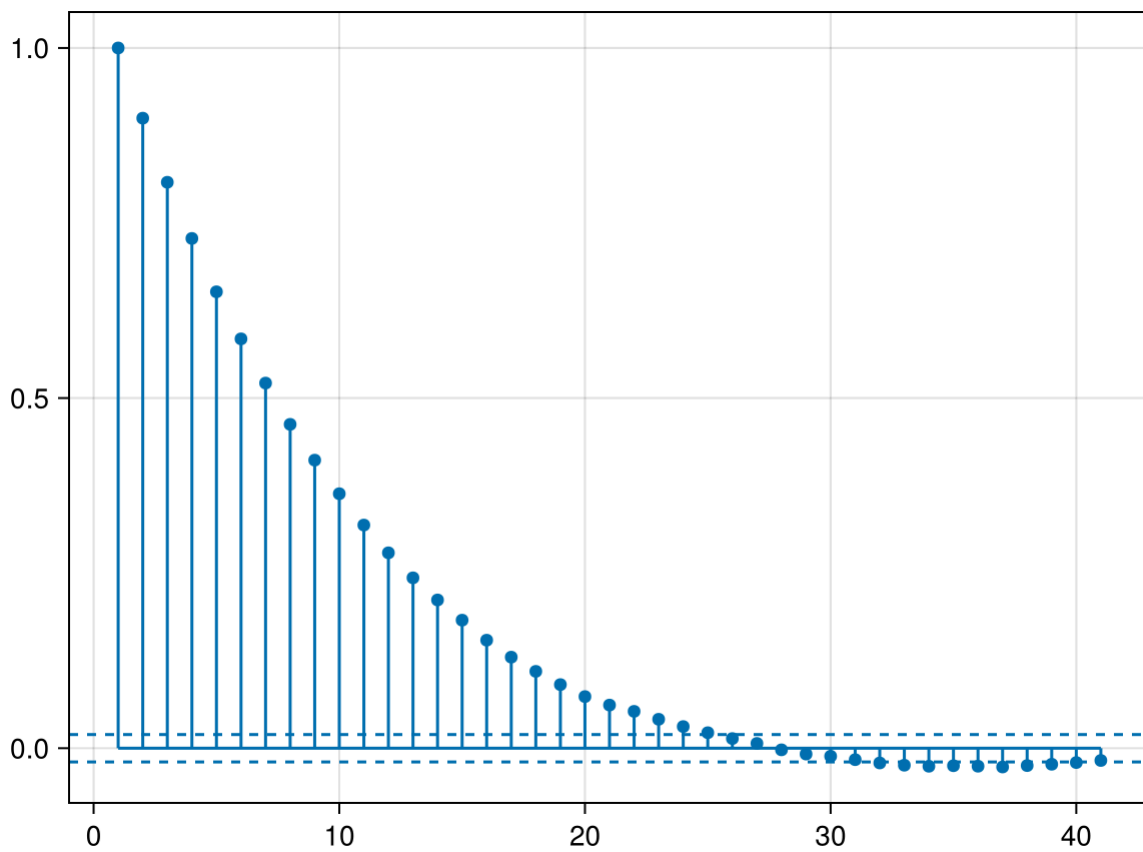
```

1 begin
2   N14 = 10000
3   alpha114 = 0.9
4
5
6   d14 = Normal()
7   sigma14 = rand(d14, N14)
8
9   x14 = zeros(N14)
10  for i in range(2,N14)
11    x14[i] = alpha114*x14[i-1]+sigma14[i]
12  end
13 end

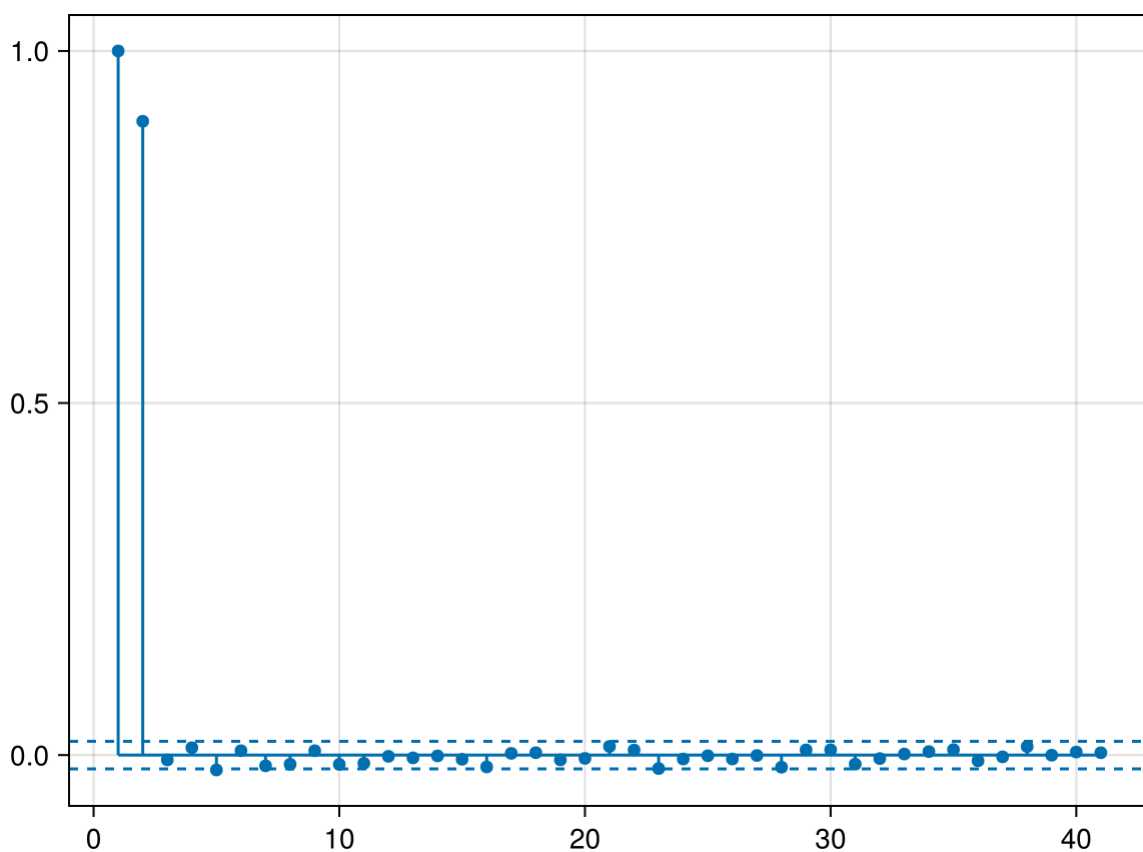
```



- Just visually looking at the plot, the time-series shows fluctuations that do not appear totally random.
- Let's analyse the ACF of the previous process:



- At variance with MA processes, a AR process shows a correlation decreases with increasing lag as a power-law.



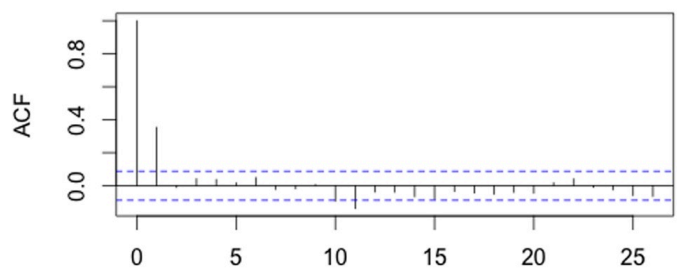
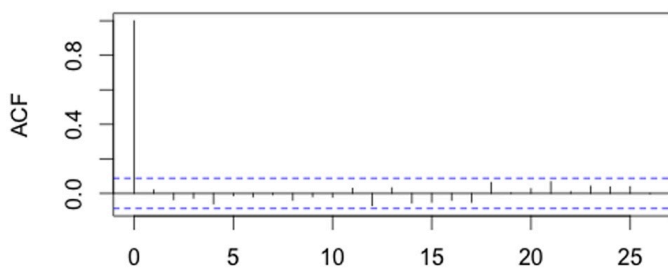
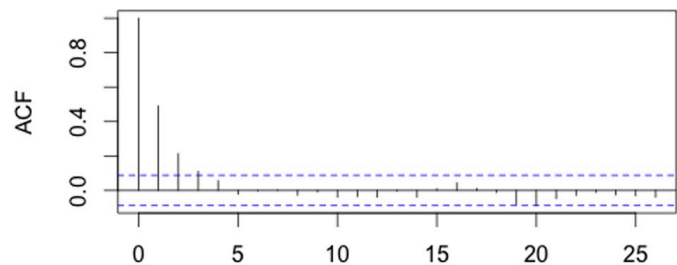
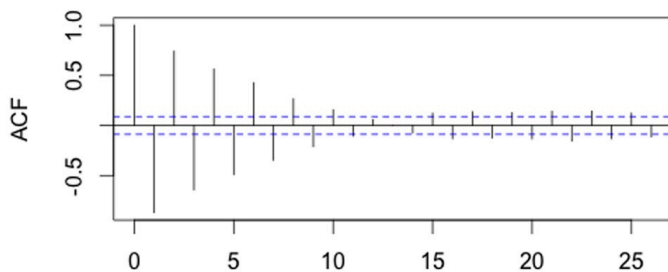
- The PACF instead shows one only lag with value significantly different from zero:

- As a matter of fact, the ACF (and PACF) are indeed a diagnostic tools to infer the nature and the order of a linear process:

Three stationary models

White noise	MA(1), any β_1	AR(1), $ \alpha_1 < 1$
$\rho(h) = 1$, when $h = 0$ $= 0$, otherwise	$\rho(h) = 1$, when $h = 0$ $= \beta_1/(1 + \beta_1^2)$, $h = 1$ $= 0$, $h \geq 2$	$\rho(h) = 1$, when $h = 0$ $= \alpha_1^h$, $h > 0$
Only lag 0 shows non-zero ACF.	Only lag 0 and 1 show non-zero ACF.	Decreasing ACF

- Try to guess the kind of process looking at the plots below:



- Upper left: The ACF shows a decreasing correlation with alternate signs. It is a MA(10) process.
- Upper right: The ACF shows a decreasing correlation as a power-law. Probably it is an AR process, although we cannot say the order. In principle it could also be a MA(4) process.
- Bottom left: The ACF shows no correlation beyond lag=0, it is a white noise process.
- Bottom right: The ACF shows a 0 correlation after lag=1. It is thus a MA(1) process, although it might also be a AR process.

As we have seen, the ACF is a powerful diagnostics but often we have ambiguous situations.

The application of the PACF will help us to solve these ambiguities, as we are going to see later.

Tip

Here we introduce an important tool!

The Autoregressive Moving Average scheme

- This approach were pioneered by Yule and Slutsky in the 1920's.
 - Yule's researches led to the notion of the autoregressive scheme.
 - Slutsky's researches led to the notion of a moving average scheme.
- We have an AutoRegressive scheme when a time series x_t is assumed to be generated as a linear function of its past values, plus a random shock:

$$x_t = \varphi_1 x_{t-1} + \dots + \varphi_p x_{t-p} + w_t$$

- Conceptually, an autoregressive scheme is one with a 'memory' in the sense that each values is correlated with p preceding values. The constants $\varphi_1, \dots, \varphi_p$ are weights measuring the influence of preceding values $x_{t-1}, \dots, x_{t-p}$ on the value x_t .
- We have a Moving Average scheme when a time series x_t is assumed to be generated as a weighted linear sum of the last $q + 1$ random shocks:

$$x_t = w_t + \theta_1 w_{t-1} + \dots + \varphi_p w_{t-q}$$

- If both schemes, the autoregressive and the moving average one, are used, we obtain the so-called Autoregressive Moving Average scheme:

$$x_t = \varphi_1 x_{t-1} + \dots + \varphi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \dots + \varphi_p w_{t-q}$$

General linear processes

- The examples we have seen so far are all part of a general family
- We call “linear process” a linear combination of noise variates w_t :

$$x_t = \sum_{i=0}^{\infty} \psi_i w_{t-i}$$

- For this kind of processes, if $\sum_{i=0}^{\infty} |\psi_i| < \infty$ then the process is stationary.
- The autocovariance of this process can be written as:

$$\rho(h) = \sigma^2 \sum_{i=0}^{\infty} \psi_{i+h} \psi_i$$

- Now we provide a set of useful definitons.

Warning:

Some useful notation. Yet, not of immediate use.

Backshift operator

- The backshift operator, **B**, is defined as:

$$Bx_t = x_{t-1}$$

- It can be extended to more powers as: $B^2x_t = (BB)x_t = B(Bx_t) = Bx_{t-1} = x_{t-2}$, so that:

$$B^kx_t = x_{t-k}$$

Difference operator

- The difference operator, ∇ , is defined as:

$$\nabla^d x_t = (1 - B)^d x_t$$

- e.g., $\nabla^1 x_t = (1 - B)^1 x_t = x_t - x_{t-1}$.
- The different operator is often used to convert non-stationary time-series to a stationary one.

Exercise: let's convert MA(1) and AR(1) processes to the general linear form

- MA(1) can be expressed as: $x_t = \beta_1 w_{t-1} + w_t = \sum_{i=0}^{\infty} \psi_i w_{t-i}$, with $\psi_0 = 1, \psi_1 = \beta_1$ and $\psi_i = 0$ for $i \geq 2$.
- AR(1) can be expressed as: $x_t = \alpha_1 x_{t-1} + w_t = w_t + \alpha_1 w_{t-1} + \alpha_1^2 w_{t-2} + \dots$ since $x_{t-1} = \alpha_1 x_{t-2} + w_{t-1}$ etc.
 - We thus have $x_t = \sum_{i=0}^{\infty} \psi_i w_{t-i}$ with $\psi_i = \alpha_1^i$.

MA(q) and AR(p)

- MA(q): $x_t = w_t + \beta_1 w_{t-1} + \beta_2 w_{t-2} + \dots + \beta_q w_{t-q}$
- AR(p): $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \dots + \alpha_p x_{t-p} + w_t$

- We can rewrite the above equations by the backshift operator:

- $$\theta(B) = 1 + \beta_1 B + \beta_2 B^2 + \dots + \beta_q B^q$$

- $$\phi(B) = 1 - \alpha_1 B - \alpha_2 B^2 - \dots - \alpha_p B^p$$

- So that:

$$\text{MA}(q) : x_t = \theta(B)w_t \quad \text{AR}(p) : \phi(B)x_t = w_t$$

- A very compact form making easier further manipulations.

ARMA(p,q) processes

- A process, x_t , is said to be an ARMA(p,q) process if it has the form:

$$\phi(B)x_t = \theta(B)w_t$$

- It is possible to prove that:
 - ARMA(p,q) is stationary if and only if the roots of $\phi(B)w$ lie outside the unit circle (i.e. $\phi(B)w \neq 0$ for all $|w| \leq 1$).
 - ARMA(p,q) is invertible if and only if the roots of $\theta(B)w$ lie outside the unit circle (i.e. $\theta(B)w \neq 0$ for all $|w| \leq 1$).
- It is also possible to convert an ARMA process, with $\phi(B)$ and $\theta(B)$ finite ordne polynomials, to infinite order MA or AR processes:
- MA: $x_t = \psi(B)w_t$, where $\psi(B) = \theta(B)/\phi(B)$.
- AR: $\pi(B)x_t = w_t$, where $\pi(B) = \phi(B)/\theta(B)$.

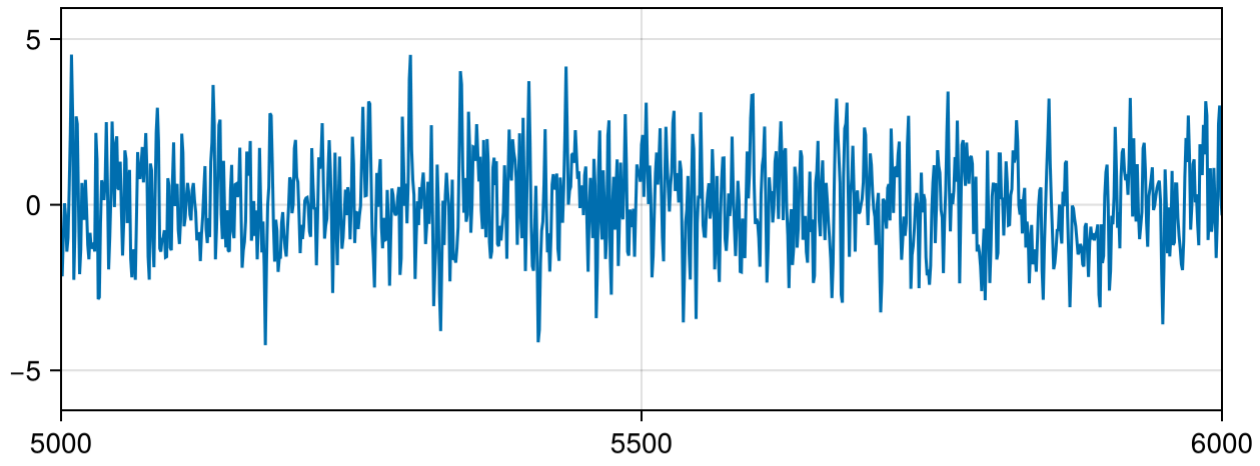
Exercize: the ACF of MA(p) and AR(q) processes

MA(1):

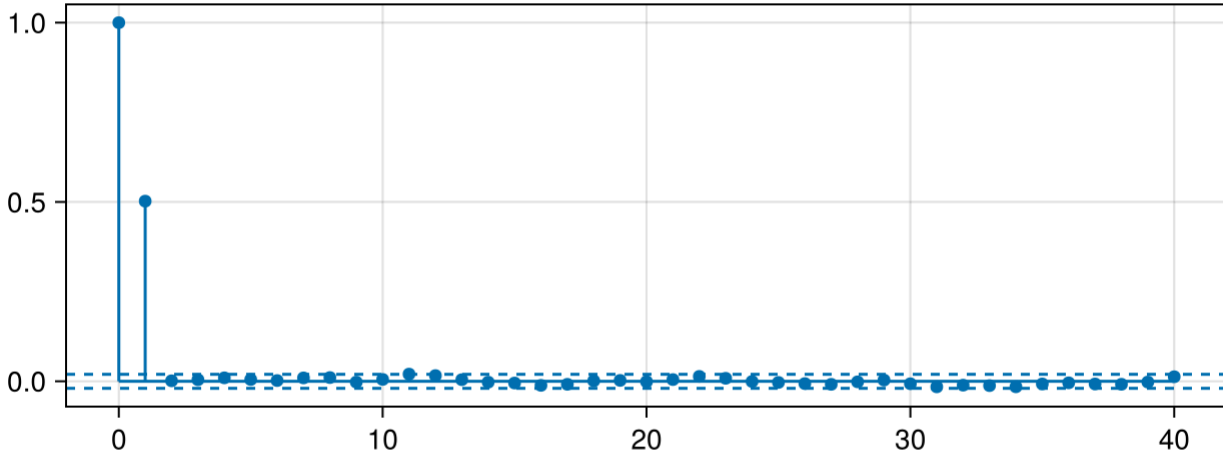
$\beta_1 =$ ▼

```
1 begin
2     N16 = 10000
3     #beta116 = 0.6
4
5
6     d16 = Normal()
7     sigma16 = rand(d16, N16)
8
9     x16 = zeros(N16)
10    for i in range(2,N16)
11        x16[i] = sigma16[i]+beta116*sigma16[i-1]
12    end
13 end
```

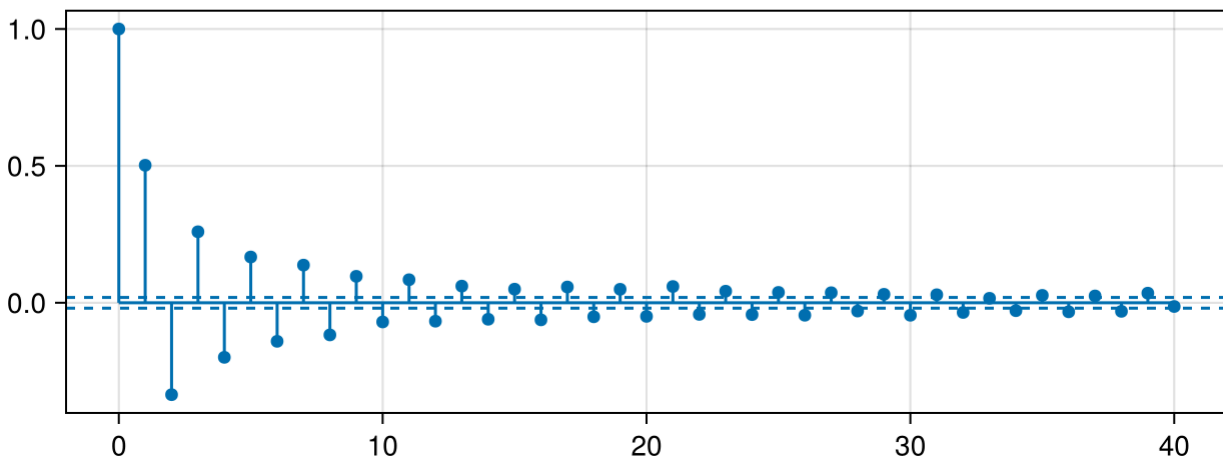

Time-Series



ACF



PACF



MA(2):

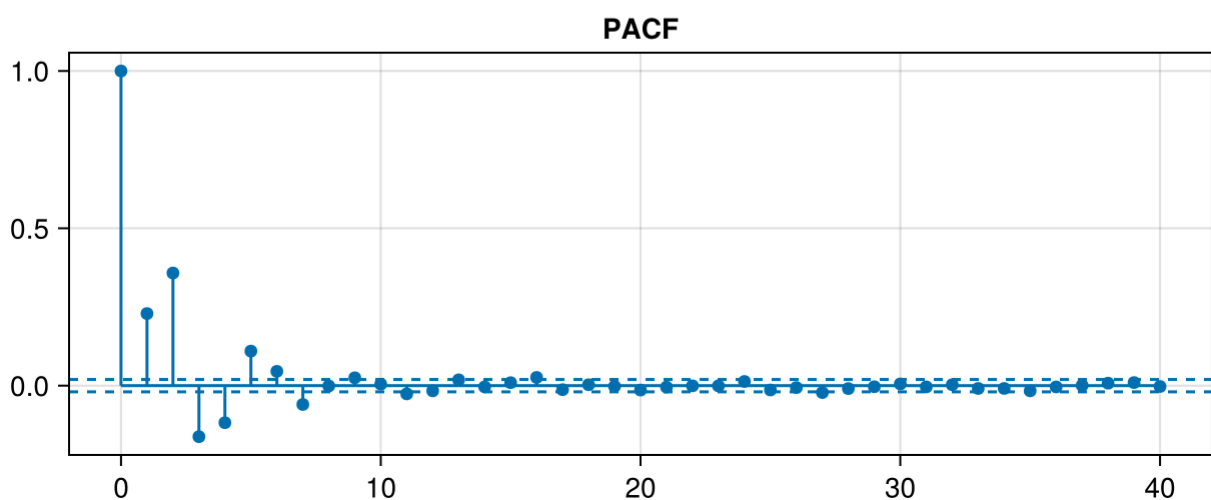
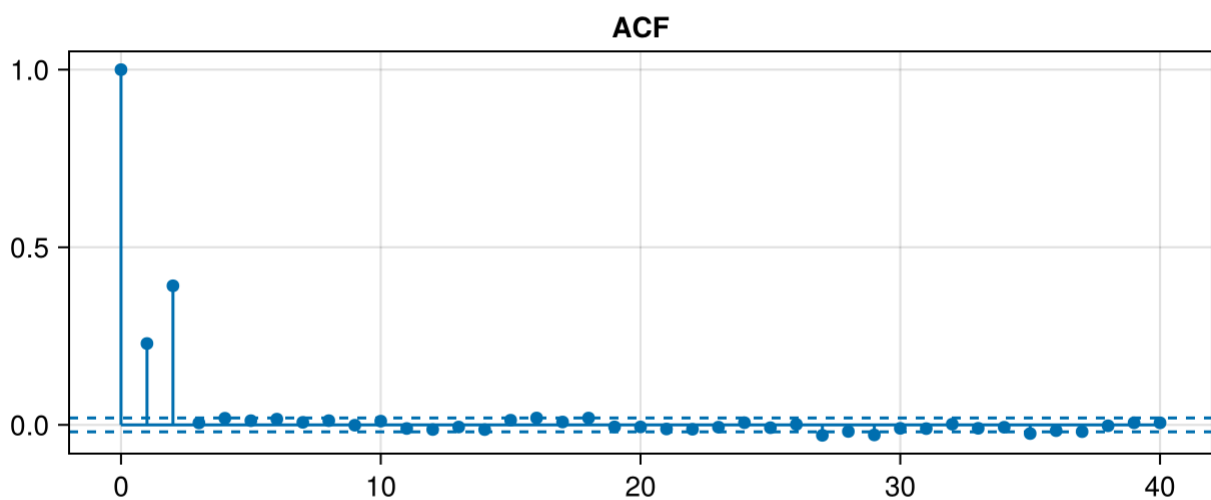
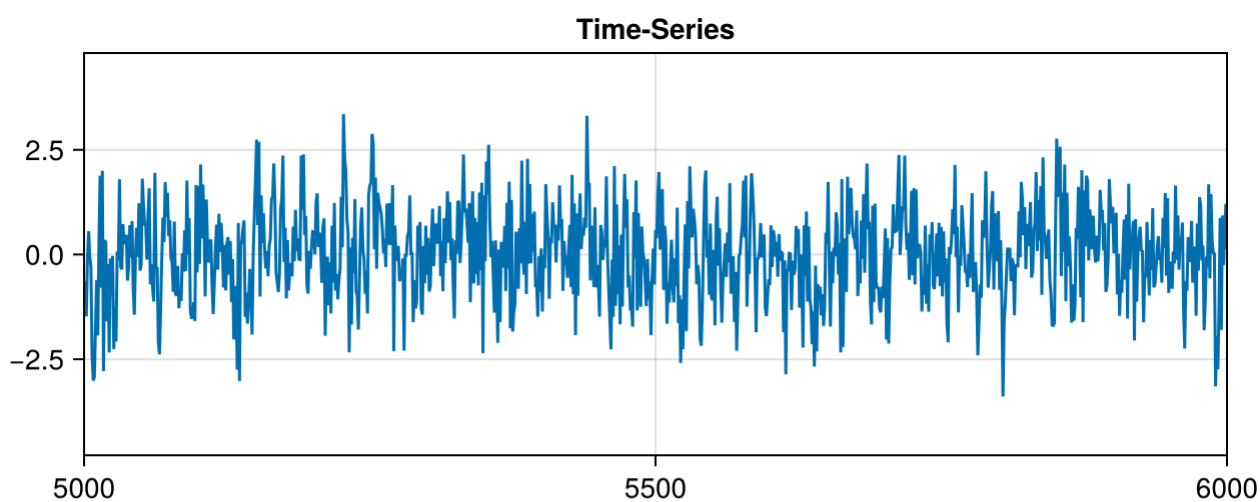
$$\beta_1 = 0.2$$

$$\beta_2 = 0.5$$

```

1 begin
2     N17 = 10000
3     beta117 = 0.2
4     beta217 = 0.5
5
6     d17 = Normal()
7     sigma17 = rand(d17, N17)
8
9     x17 = zeros(N17)
10    for i in range(3,N17)
11        x17[i] = sigma17[i]+beta117*sigma17[i-1]+beta217*sigma17[i-2]
12    end
13 end

```

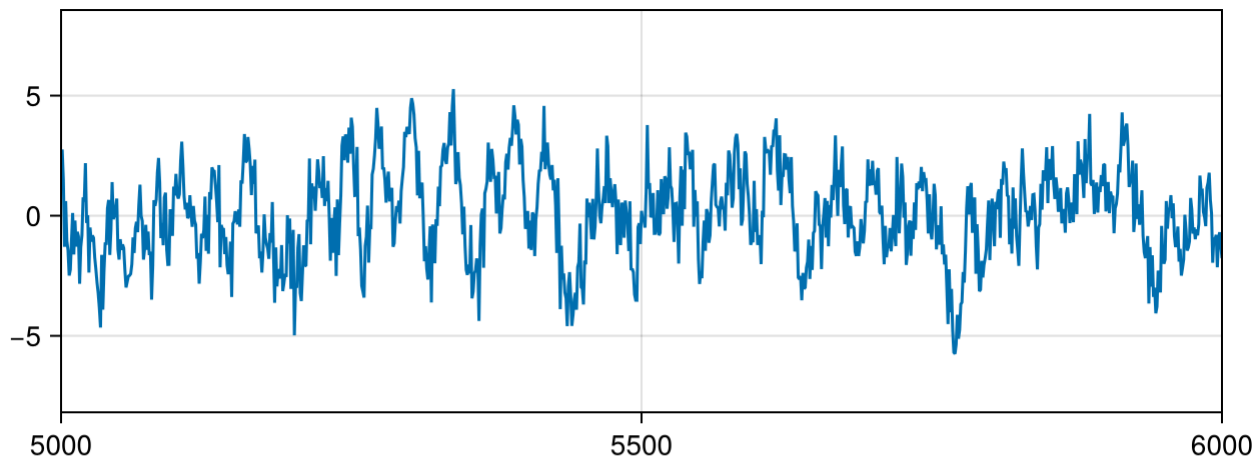


MA(10):

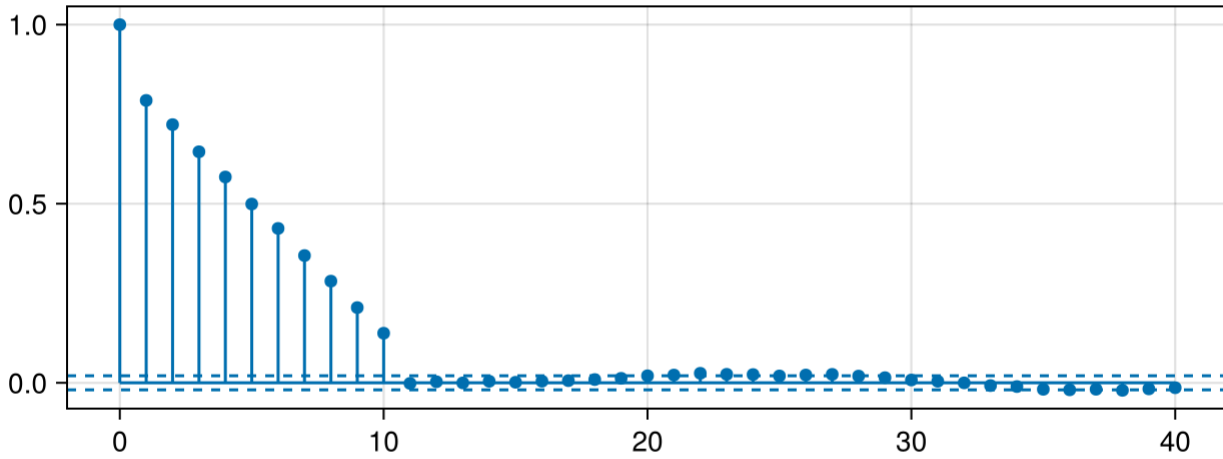
$\beta_j = 0.5$ for $j = 1 \dots 10$.

```
1 begin
2     N19 = 10000
3     beta19 = 0.5
4
5     d19 = Normal()
6     sigma19 = rand(d19, N19)
7
8     x19 = []
9
10    x19 = zeros(10)
11    for i in range(11,N19)
12        xt = sigma19[i]
13        for j in range(1,10)
14            xt = xt + beta19*sigma19[i-j]
15        end
16        push!(x19,xt)
17    end
18 end
```

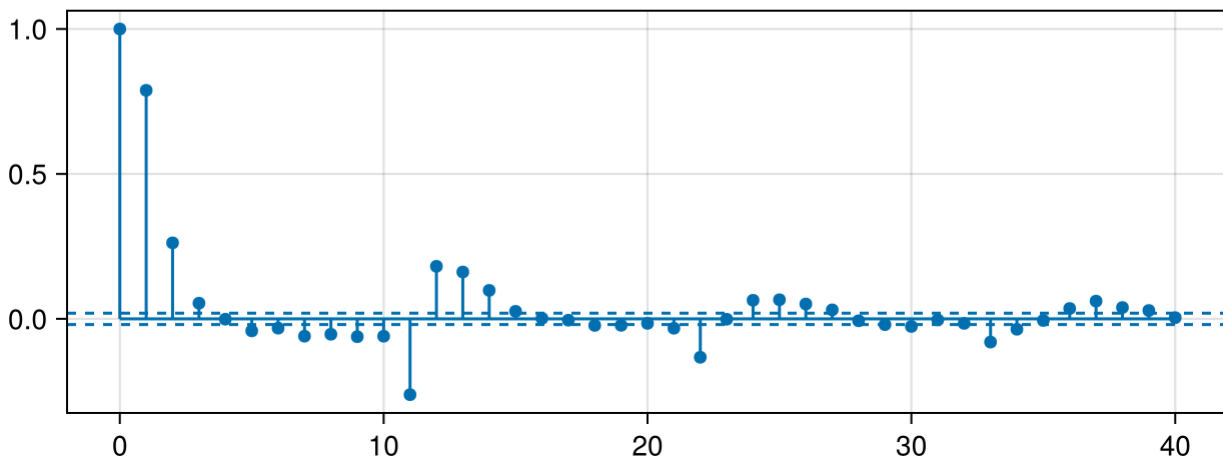
Time-Series



ACF



PACF



- AR(1): $\alpha_1 = 1/2$.

AR(1):

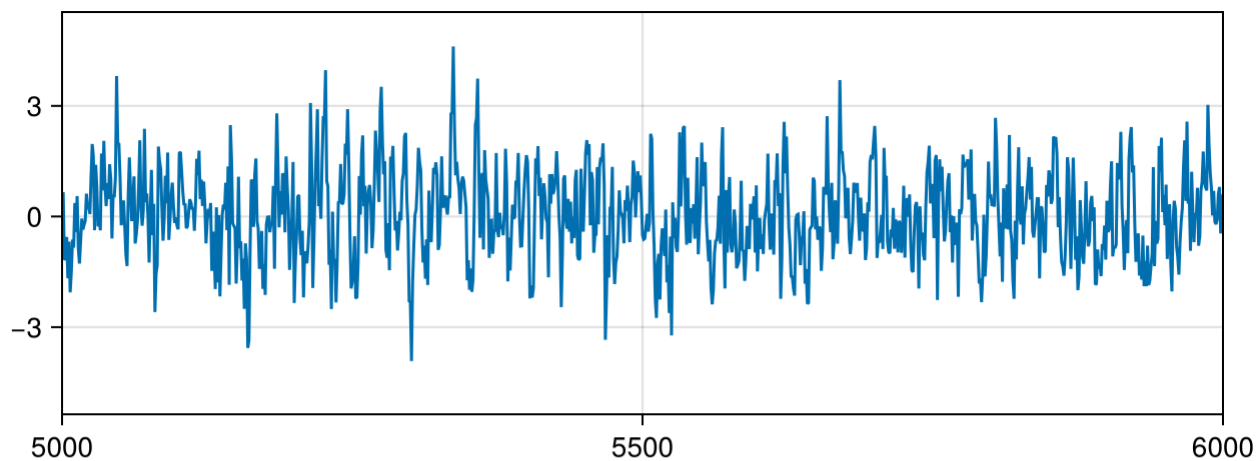
$\alpha_1 =$ ▼

```

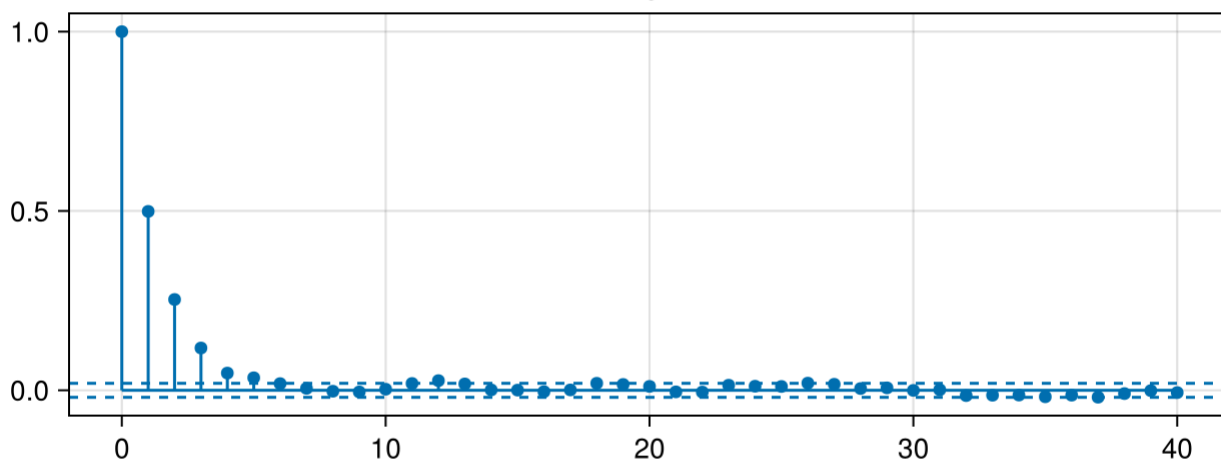
1 begin
2     N20 = 10000
3
4
5     d20 = Normal()
6     sigma20 = rand(d20, N20)
7
8     x20 = [0.,]
9
10    for i in range(2,N20)
11        push!(x20,alpha20*x20[i-1]+sigma20[i])
12    end
13 end

```

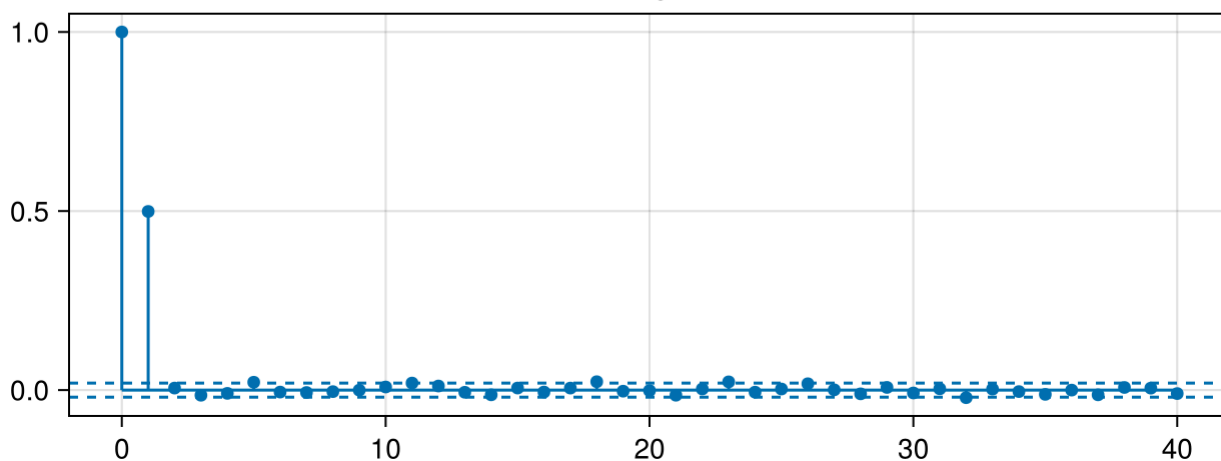
Time-Series



ACF



PACF



AR(2):

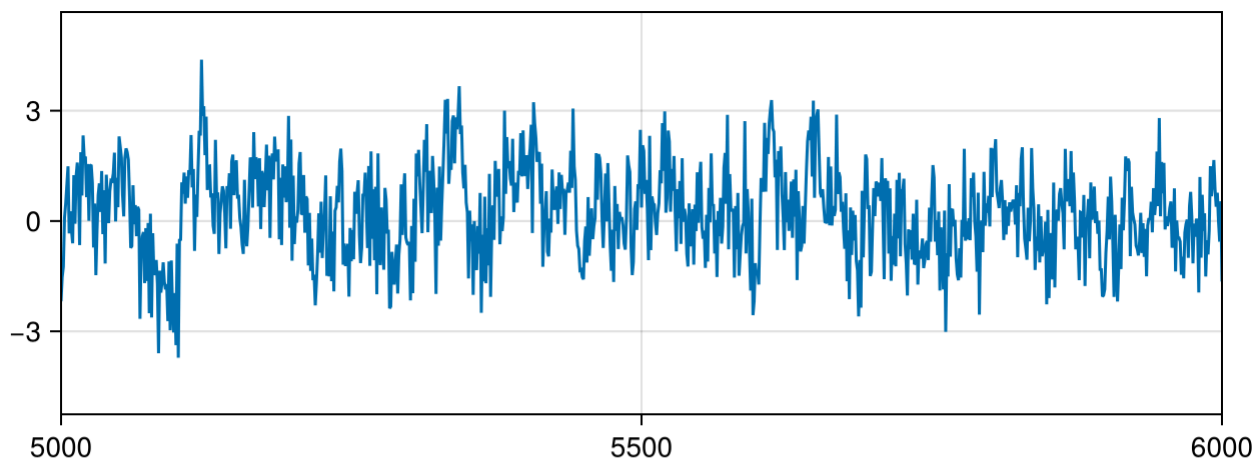
$$\alpha_1 = 0.2 \quad \alpha_2 = 0.5$$

```

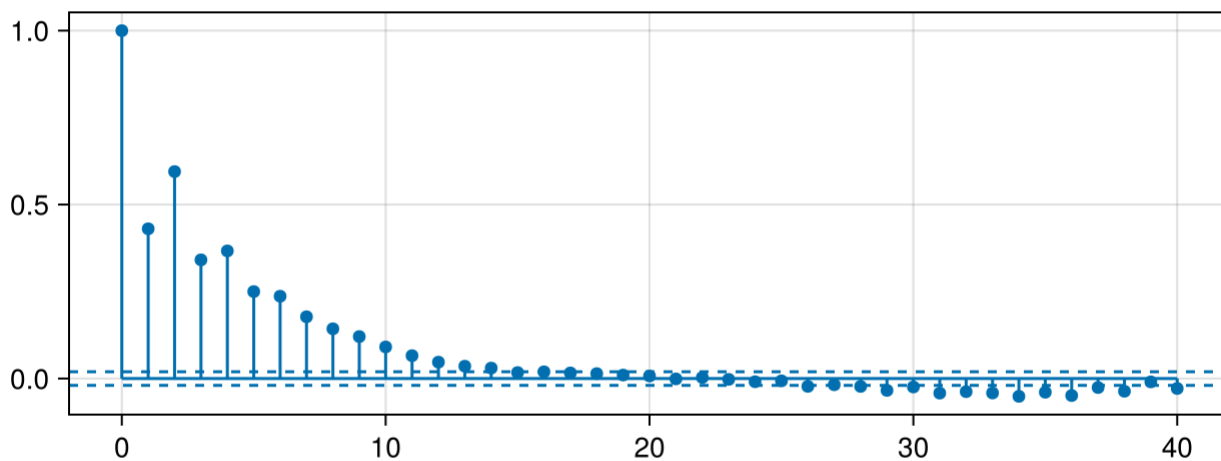
1  begin
2      N21 = 10000
3      alpha121 = 0.2
4      alpha221 = 0.5
5
6      d21 = Normal()
7      sigma21 = rand(d21, N21)
8
9      x21 = [0.,0.]
10
11     for i in range(3,N21)
12         push!(x21,alpha121*x21[i-1]+alpha221*x21[i-2]+sigma21[i])
13     end
14 end

```

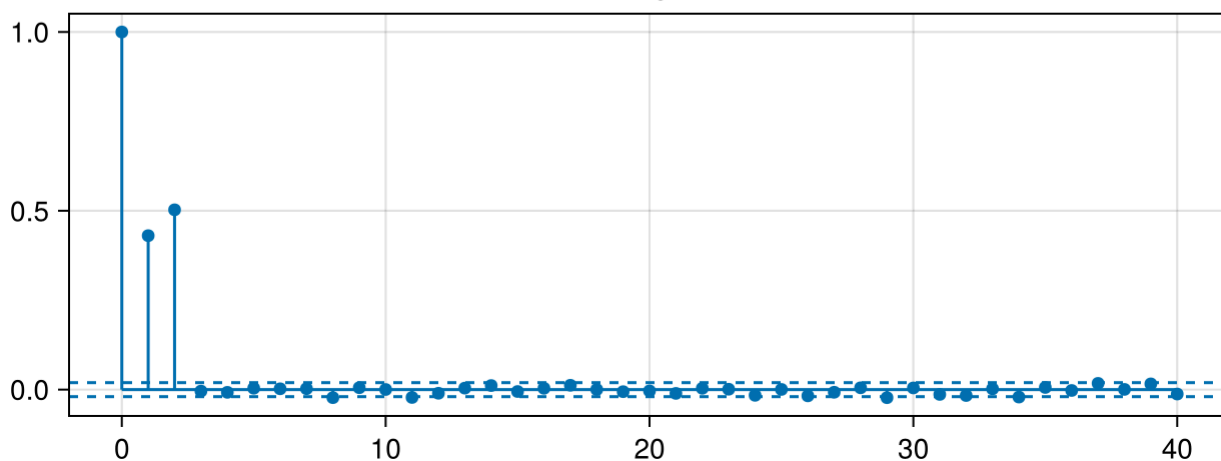
Time-Series



ACF



PACF

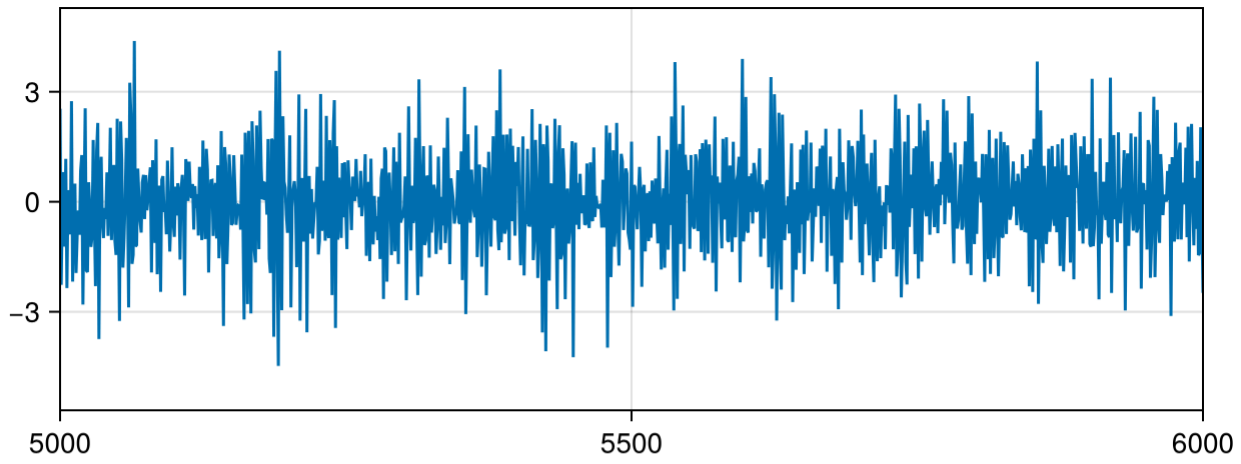


AR(5):

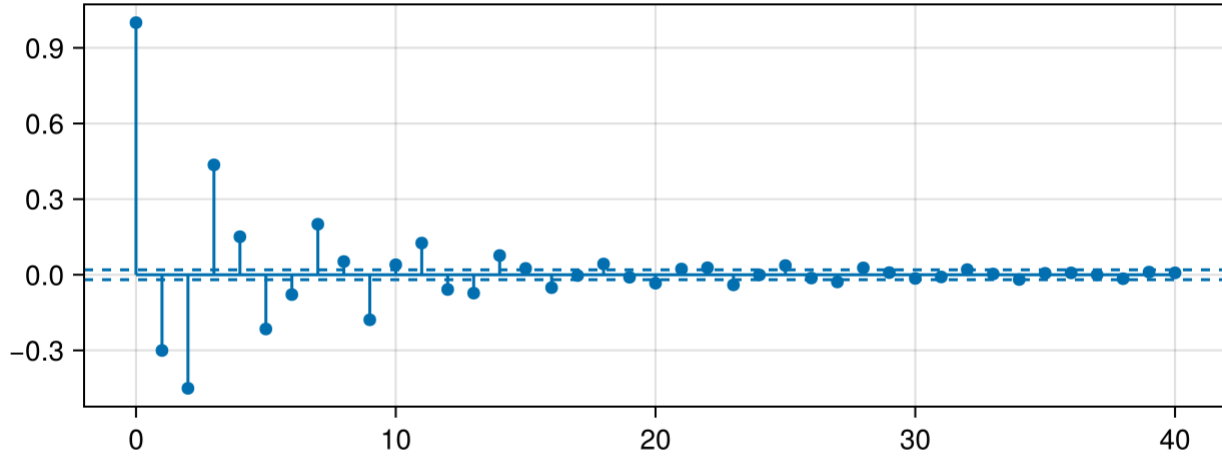
$$\alpha_{12} = -0.5 \quad \alpha_{345} = 0.3$$

```
1 begin
2   N22 = 10000
3   alpha1222 = -0.5
4   alpha34522 = 0.3
5
6
7   d22 = Normal()
8   sigma22 = rand(d22, N22)
9
10  x22 = [0.,0.,0.,0.,0.]
11
12  for i in range(6,N22)
13    push!(x22,alpha1222*x22[i-1]+alpha1222*x22[i-2]+alpha34522*x22[i-
14    3]+alpha34522*x22[i-4]+alpha34522*x22[i-5]+sigma22[i])
15  end
16 end
```

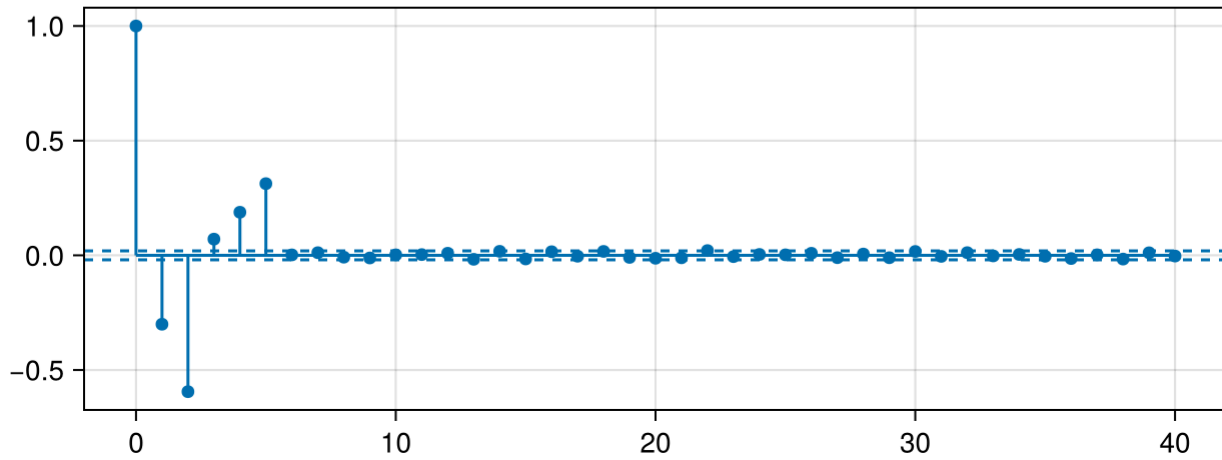
Time-Series



ACF



PACF



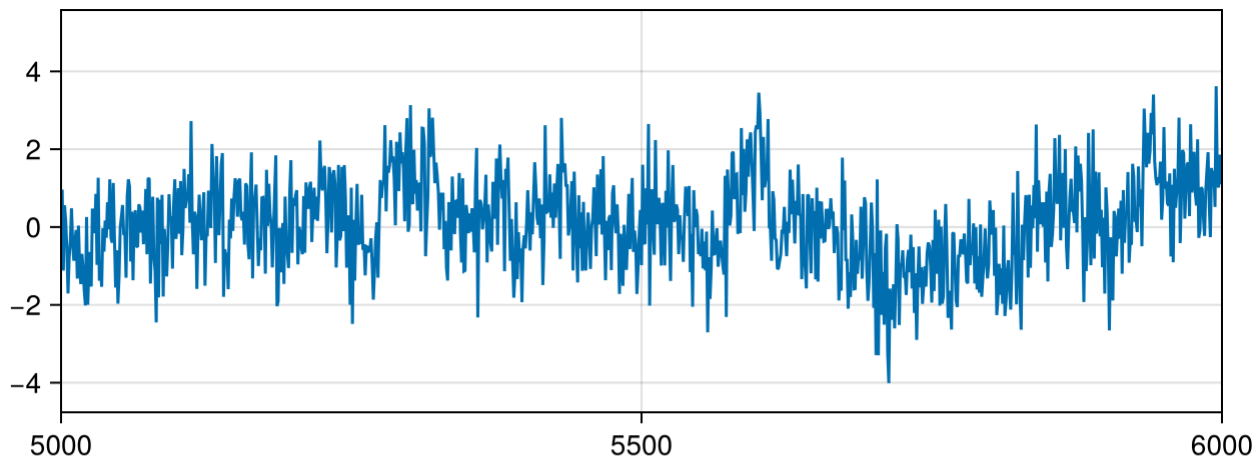
- AR(8), $\alpha_j = 1/9$ for $j = 1..8$.

AR(8):

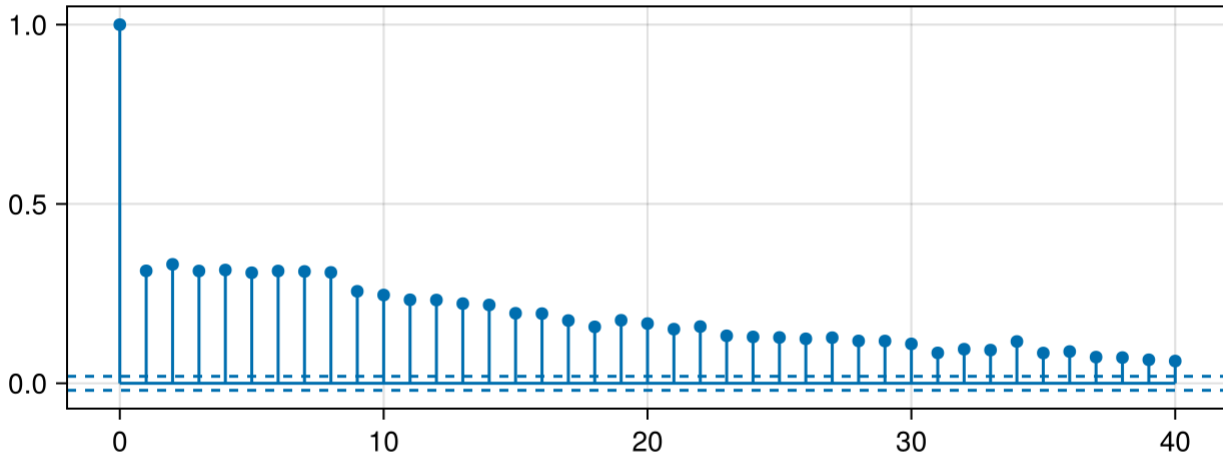
$\alpha_j = 0.1$ for $j = 1..8$.


```
1 begin
2   N23 = 10000
3   alpha23 = 0.1
4
5
6   d23 = Normal()
7   sigma23 = rand(d23, N23)
8
9   x23 = [0.,0.,0.,0.,0.,0.,0.,0.,0.]
10
11  for i in range(9,N23)
12    xt = sigma23[i]
13    for j in range(1,8)
14      xt = xt + alpha23*x23[i-j]
15    end
16    push!(x23,xt)
17  end
18 end
```

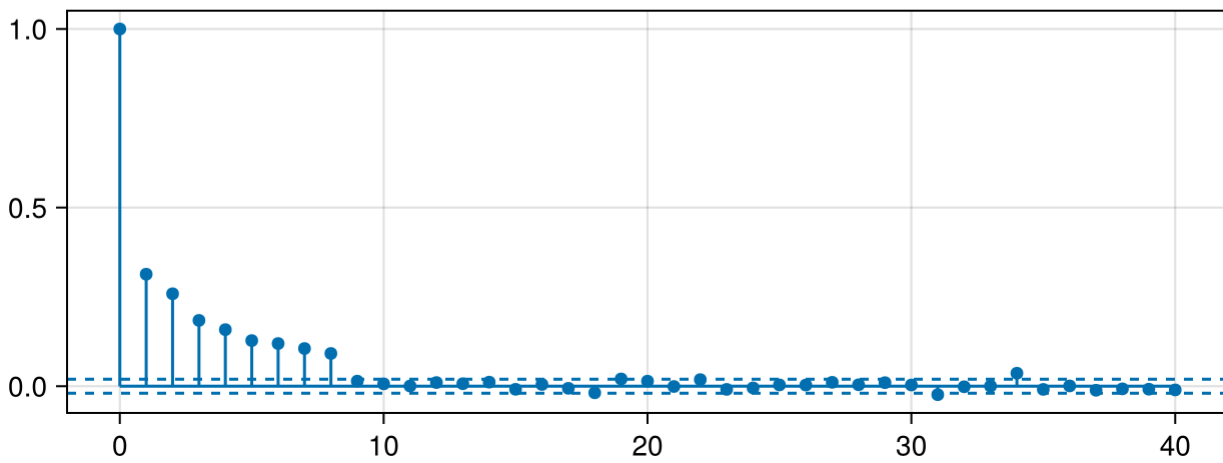
Time-Series



ACF



PACF



- ARMA(1,1): $\alpha_1 = 1/2, \beta_1 = 1/2$.

ARMA(1,1):

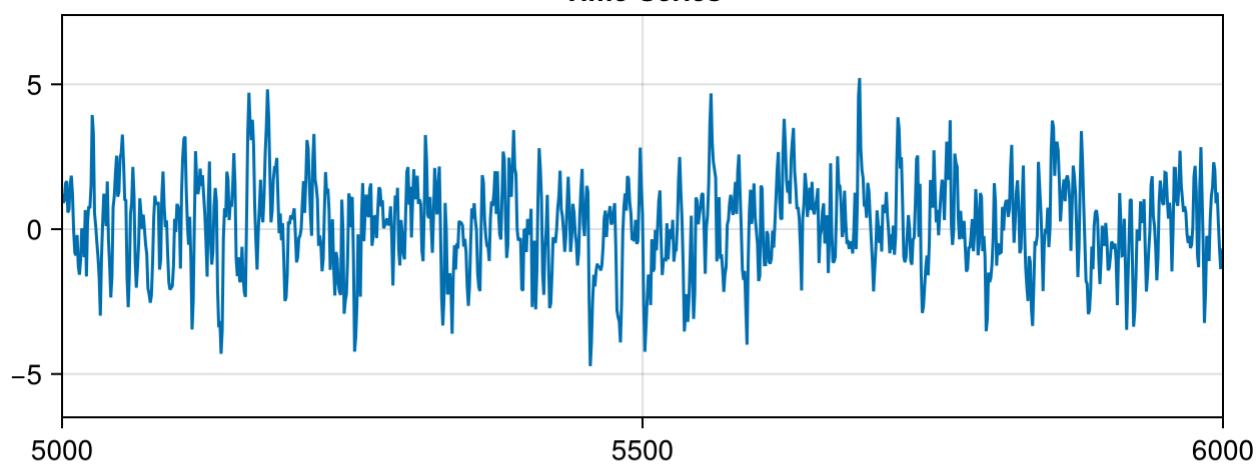
$\alpha_1 =$ $\beta_1 =$

```

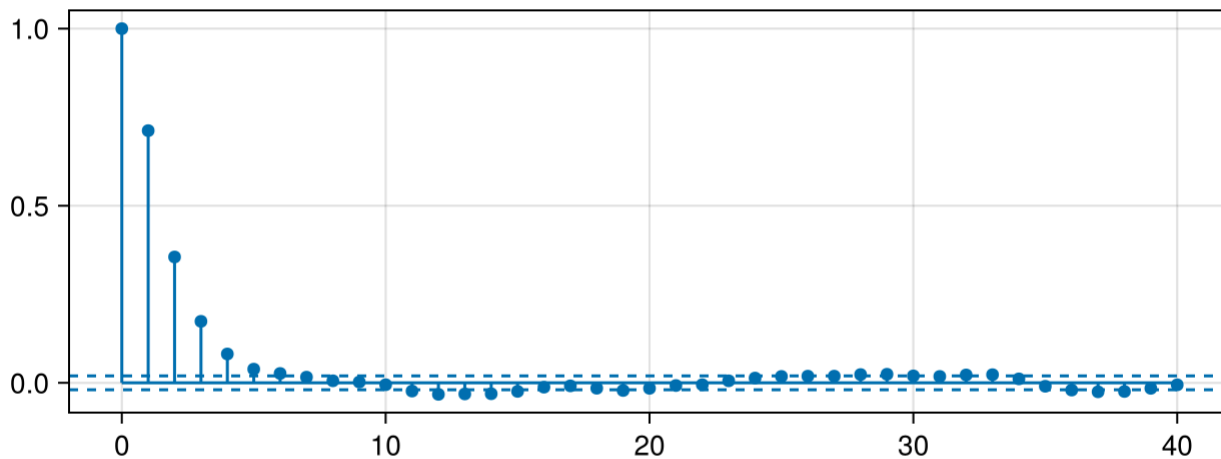
1 begin
2     N24 = 10000
3
4
5     d24 = Normal()
6     sigma24 = rand(d24, N24)
7
8     x24 = [0.,]
9
10    for i in range(2,N24)
11        xt = sigma24[i] + alpha124*x24[i-1] + beta124*sigma24[i-1]
12        push!(x24,xt)
13    end
14 end

```

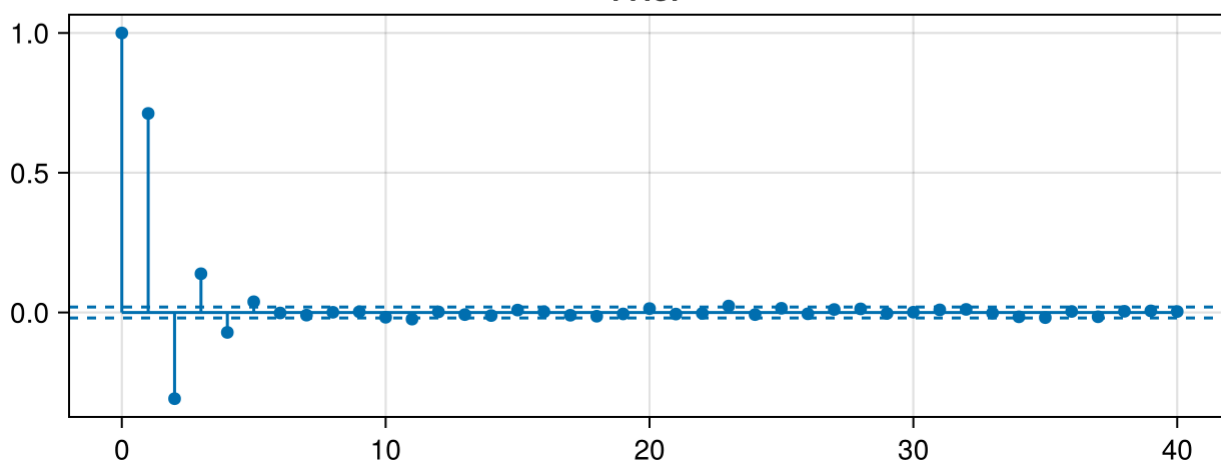
Time-Series



ACF



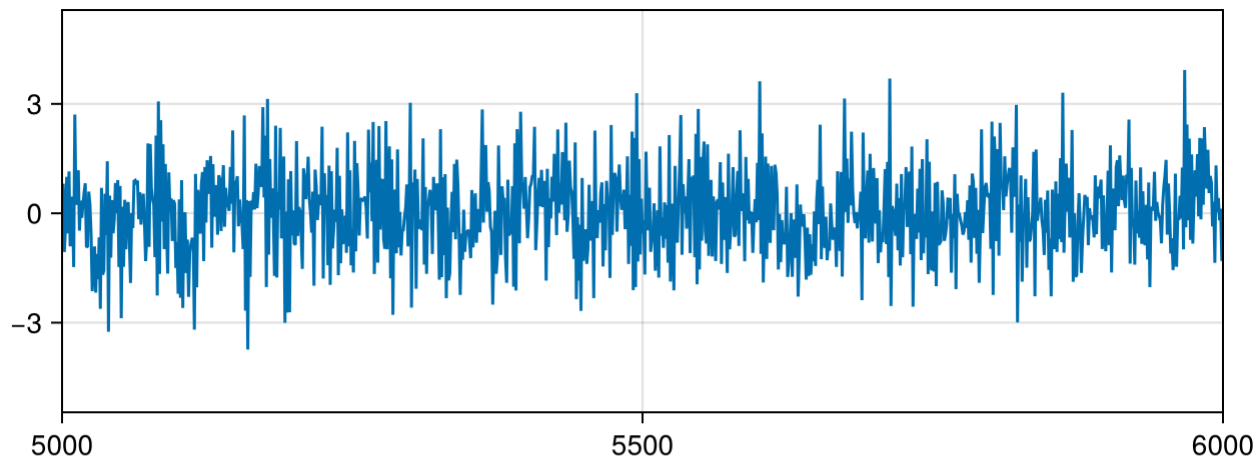
PACF



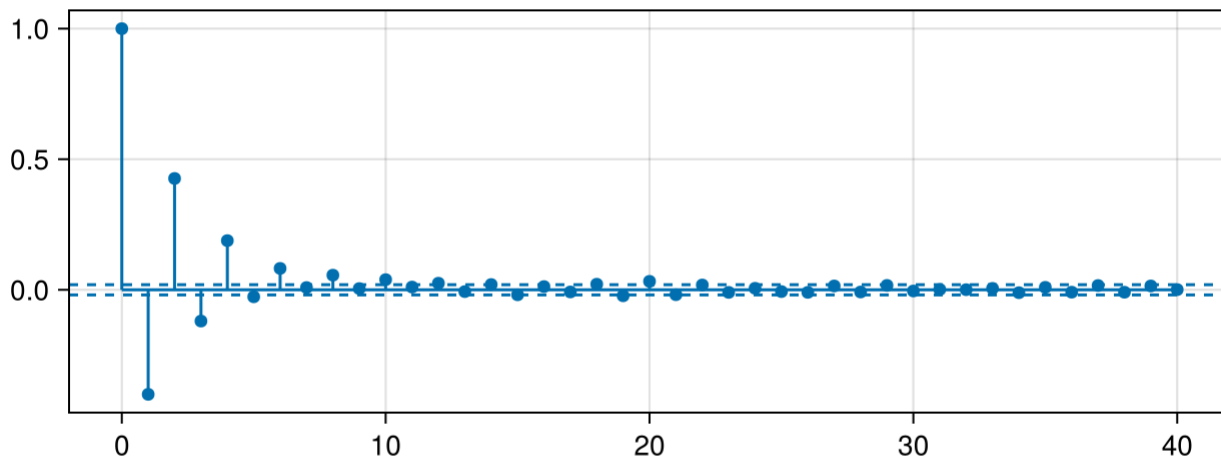
- ARMA(2,1): $\alpha_1 = 1/6, \alpha_2 = 1/2, \beta_1 = 1/2$.

```
1 arma_rvs25 = arma(10000, 1., SVector(1/6,0.5),SVector(0.5));
```

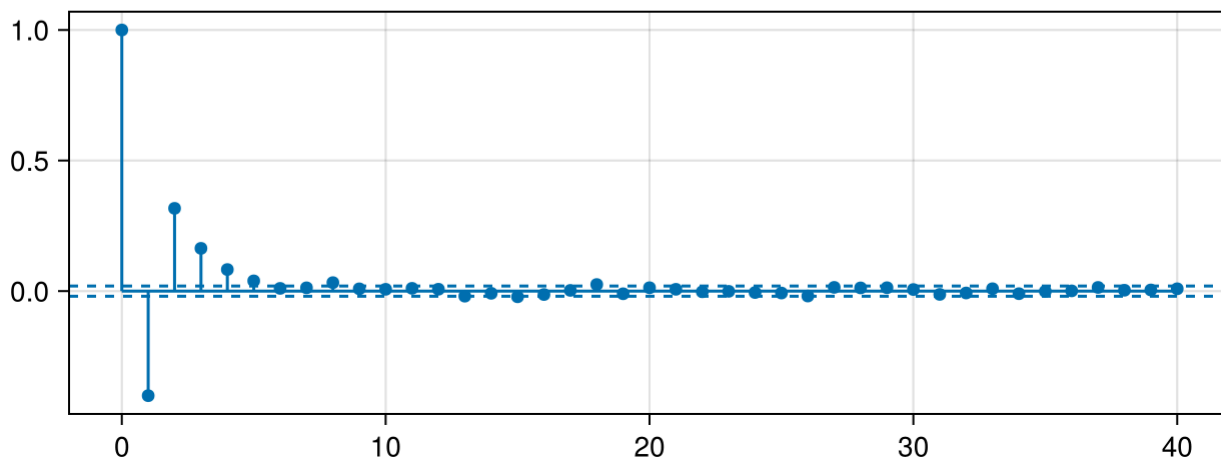
Time-Series



ACF



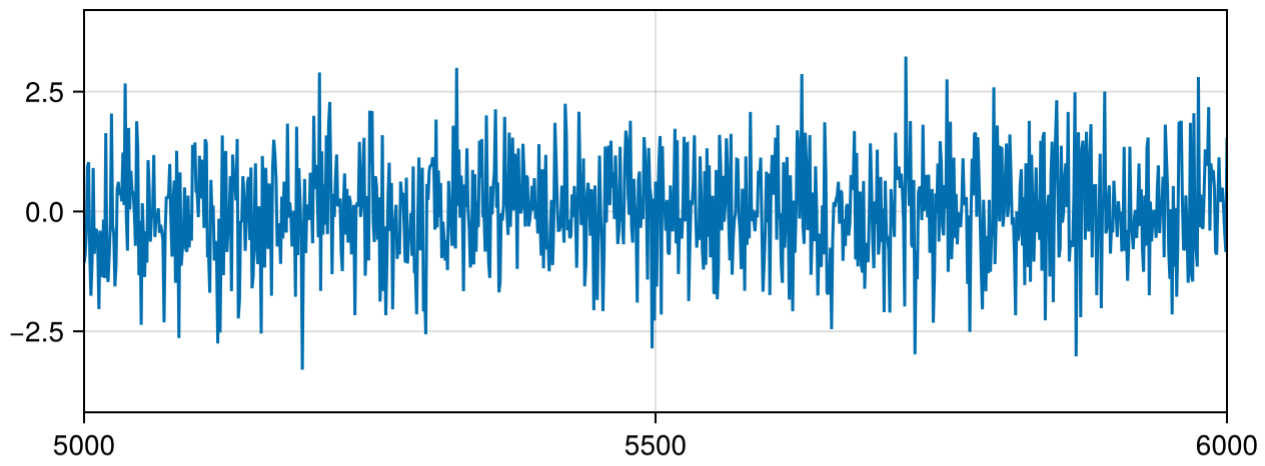
PACF



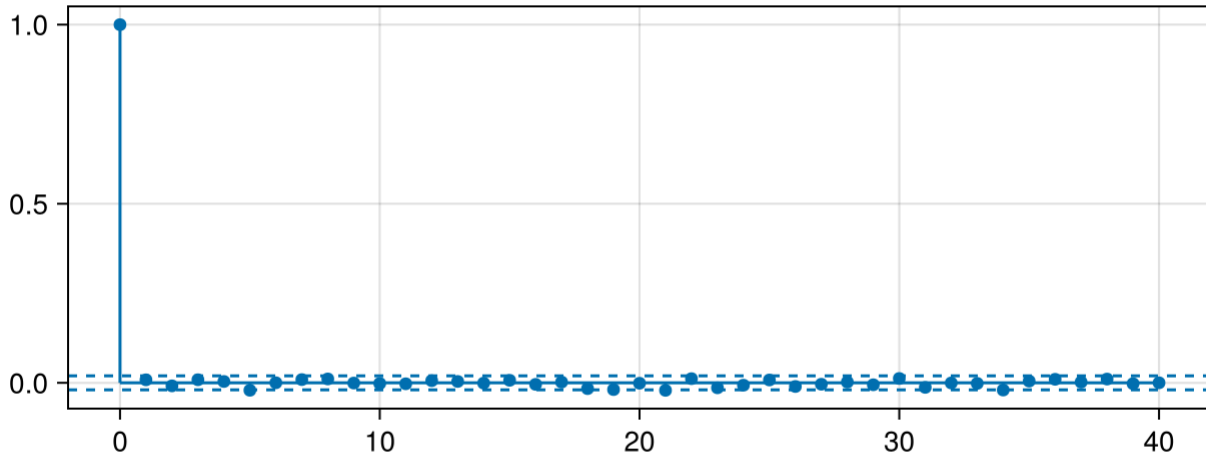
- ARMA(2,2): $\alpha_1 = -1/2, \alpha_2 = 1/4, \beta_1 = -1/2, \beta_2 = 1/4$.

```
1 arma_rvs26 = arma(10000, 1., SVector(-0.5,0.25),SVector(-0.5,0.25));
```

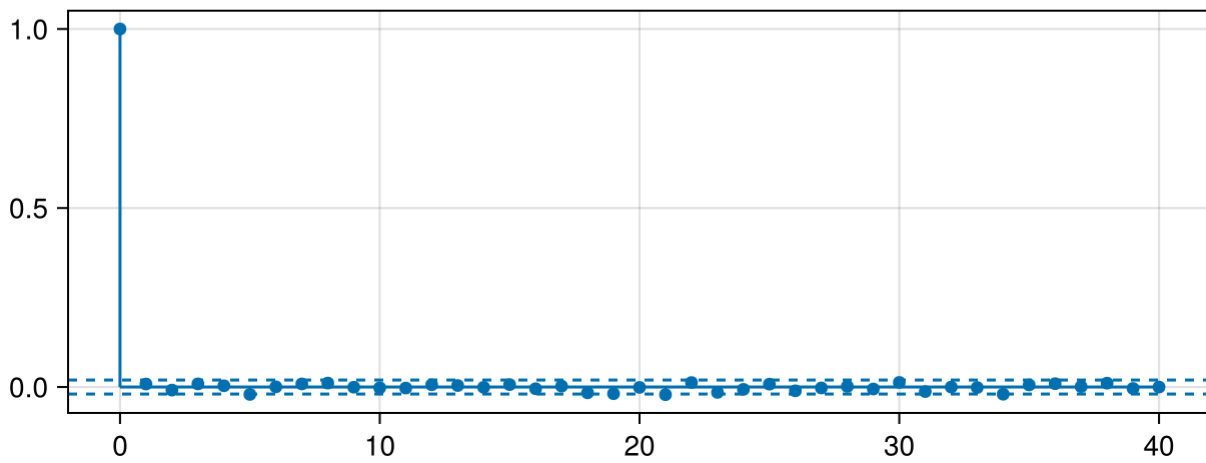
Time-Series



ACF



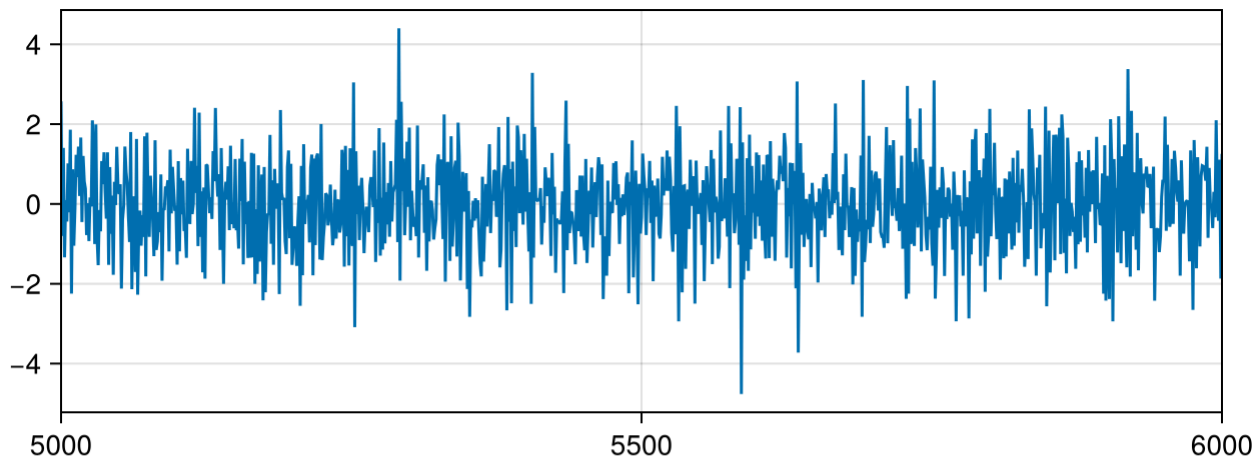
PACF



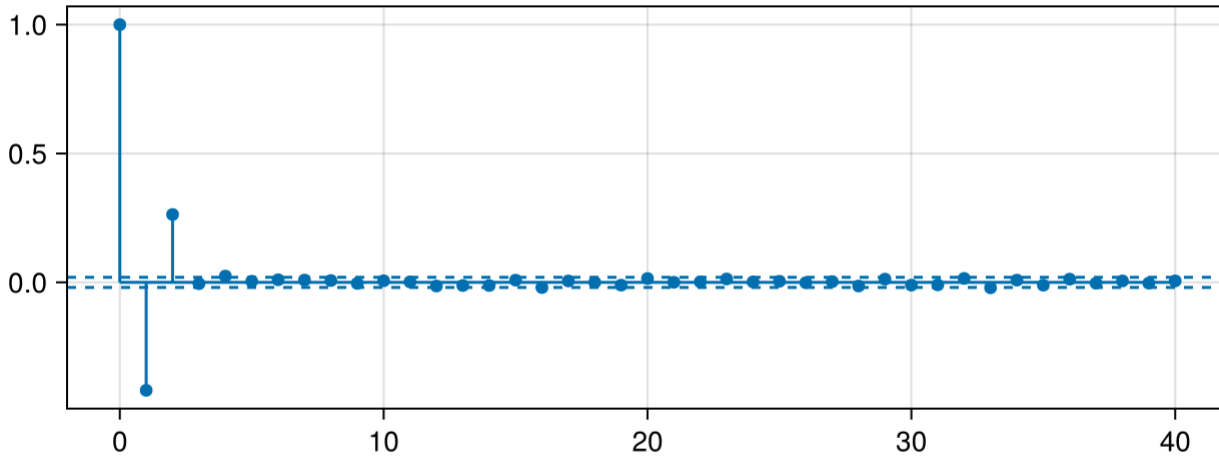
- ARMA(2,2): $\alpha_1 = 1/9, \alpha_2 = 1/9, \beta_1 = 1/2, \beta_2 = -1/4$.

```
1 arma_rvs27 = arma(10000, 1., SVector(1/9,1/9),SVector(0.5,-0.25));
```

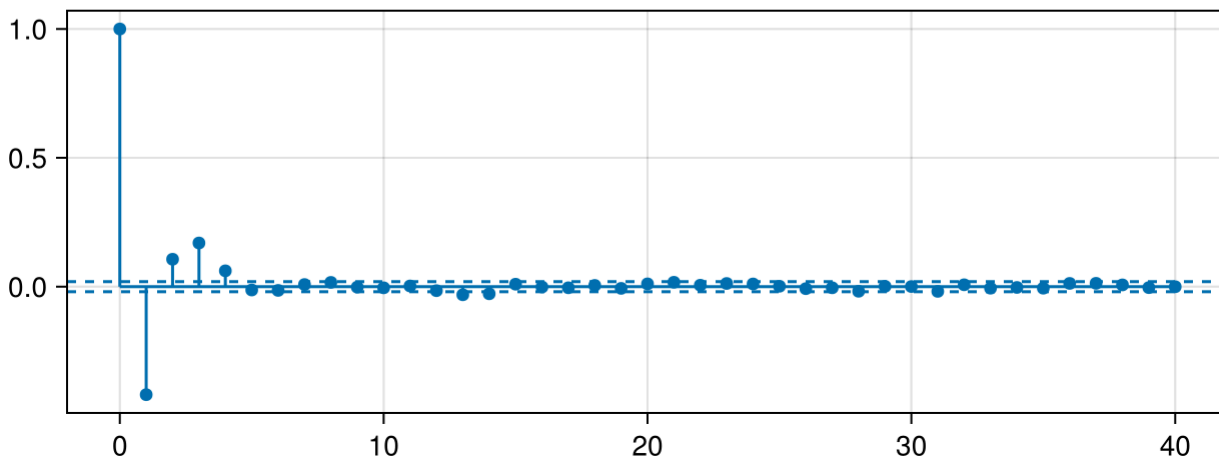
Time-Series



ACF



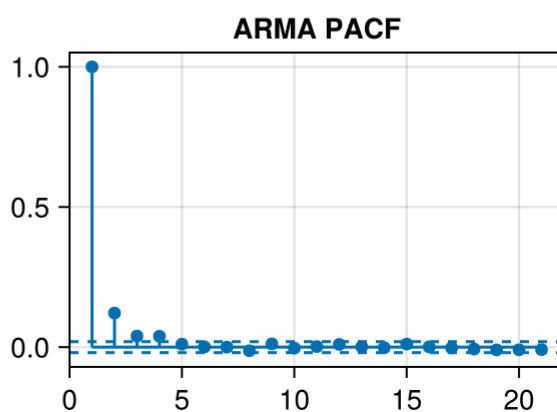
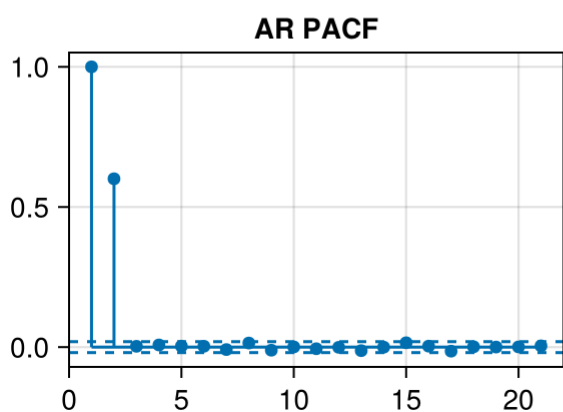
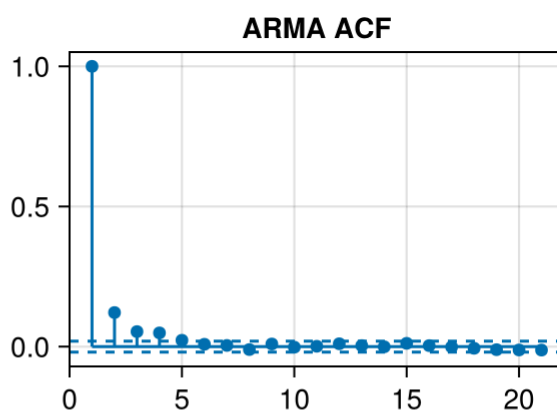
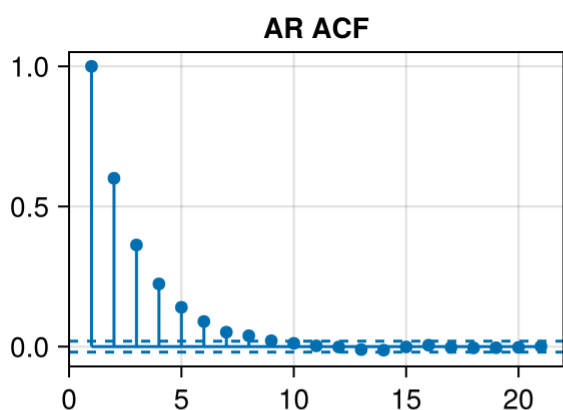
PACF



- The ACF is a powerful method to derive the order of the process but as soon as the processes become complex cannot give unambiguous answers.
- A combined use of ACF and PACF can solve more cases.

	MA(q)	AR(p)	ARMA(p, q)
ACF	zero lags > q	tails off	tails off
PACF	tails off	zero lags > p	tails off

- For instance, let's compare AR(1) with $\alpha_1 = 0.6$ to ARMA(1,1) with $\alpha_1 = 1/2, \beta_1 = 2/5$.



ARIMA(p,d,q)

- Autoregressive Integrated Moving Average
- A process $\mathbf{x}_t$ is ARIMA(p,d,q) if $\mathbf{x}_t$, differenced “d times” ($\nabla^d \mathbf{x}_t$), is an ARMA(p,q) process.

$$\phi(B)\nabla^d \mathbf{x}_t = \theta(B)w_t \quad \phi(B)(1 - B)^d \mathbf{x}_t = \theta(B)w_t$$

- Having chosen the right orders of the moving average autoregressive processes there are different strategies for determining the ARIMA parameters (Yule-Walker equations, maximum-likelihood, etc.).
- Typically, one could follow a procedure known as Box-Jenkins method that, with some simplification, essentially is:
 1. Plot the data
 2. Difference until series is stationary (find d)
 3. Examine ACF and PACF to guess p and q
 4. Fit ARIMA(p,d,q) to the original data
 5. Check model diagnostics

Nowdays, with modern computers, a grid-search (or something more elaborated and effective) is often a feasible solution.

- While the standard ARMA and ARIMA models are the workhorses of time series analysis, real-world data is rarely “standard.” It often contains seasonal patterns, long-term dependencies, or external influences that basic models simply aren’t equipped to handle.
- To tackle these complexities, extended variants of the basic models designed to capture specific data behaviors do exist.

SARIMA (Seasonal ARIMA)

- Standard ARIMA models are great at handling trends, but they fail with repeating patterns. SARIMA adds a seasonal component to the mix. It treats the seasonal part of the data as its own ARIMA process.
- These models are often expressed as $ARIMA(p, d, q) \times (P, D, Q)_s$, where the uppercase letters represent the seasonal orders and s is the number of periods per season.

ARIMAX (ARIMA with Exogenous Variables)

- Sometimes, the future of a variable isn’t just dictated by its own past, but by outside factors. The X stands for Exogenous. This model includes external covariates as predictors alongside the internal lags of the time series.

FARIMA / ARFIMA (Fractionally Integrated ARIMA)

- In a standard ARIMA model, the integration parameter (d) is an integer. However, some processes exhibit Long Memory, where the impact of a past shock decays very slowly over time. FARIMA allows d to be a non-integer (a fraction) to take care of these cases.

CARMA (Continuous models)

- While the models we’ve discussed so far (like SARIMA or FARIMA) operate on discrete regularly sampled time steps, the CARMA (Continuous-time Auto-Regressive Moving Average) are designed for data that exists in a continuous flow, even if our observations of it are irregular or missing.

ARCH models

- Another form of nonstationarity occurs when the variance, rather than the local mean level, of the time series changes during the observation. This is commonly called *volatility*, in econometric contexts.
- These kind of problems can be addressed by the autoregressive conditional heteroscedastic (ARCH) models. Here the variance is assumed to be a stochastic autoregressive process depending on previous values.
- For instance, in a ARCH(1) model:

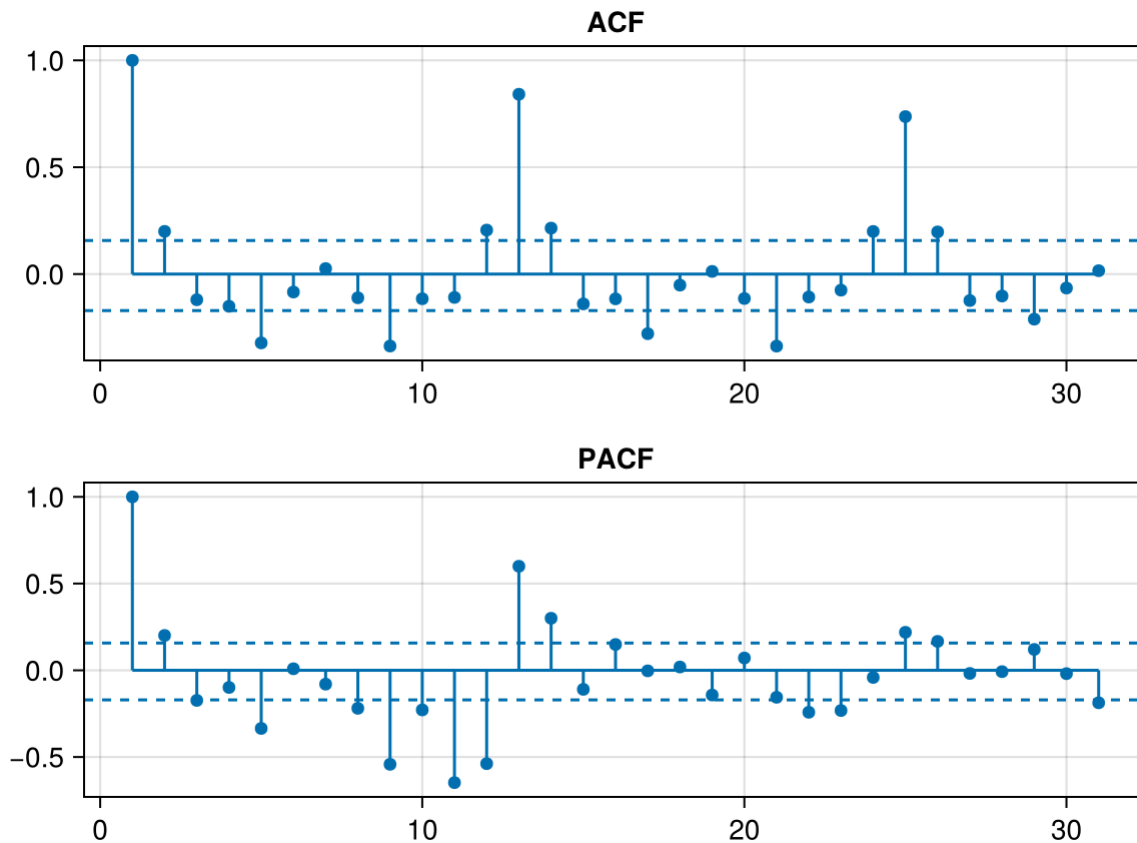
$$\text{Var}_{\text{ARCH}(1)}(\epsilon_i) = w_t + \alpha_1 \text{Var}(\epsilon_{i-1})$$

Tip

We go back to the airline passenger dataset

Exercise: fit a dataset by an ARIMA model

- Let's go back to the airline passenger dataset.
- Converting to log and differencing we got a stationary time-series getting rid of the trend.
- We also have identified a 12 month seasonal component.
- Let's study the ACF and PACF of the differenced (lag 1) time-series.



- The ACF becomes 0 after 1 lag, yet the seasonality is clearly visible, and the same is true for the PACF.
- Diagnostic plots of the time series can be used along with heuristic rules to determine the hyperparameters of the ARIMA model. These are occasionally good, but this case require a different approach.
- It is indeed an optimization problem. We deal with it finding the best p and q both for the 12 month seasonal component and the residuals.
- For interested reader, the optimization algorithm is described in, e.g., [Durbin & Siem Jan \(2012\)](#).

```
1 modelar = auto_arima(train[!,"Passengers_log"]; seasonal = 12, allow_mean=true);
```

Best result: $p = 1$, $q = 1$ for the residuals, while the seasonal component is modeled with $p = 0$, $q = 1$ and 1 differentiation(s).

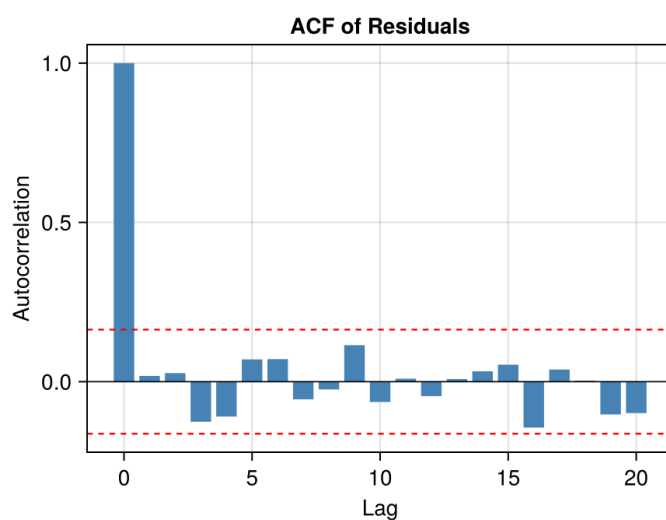
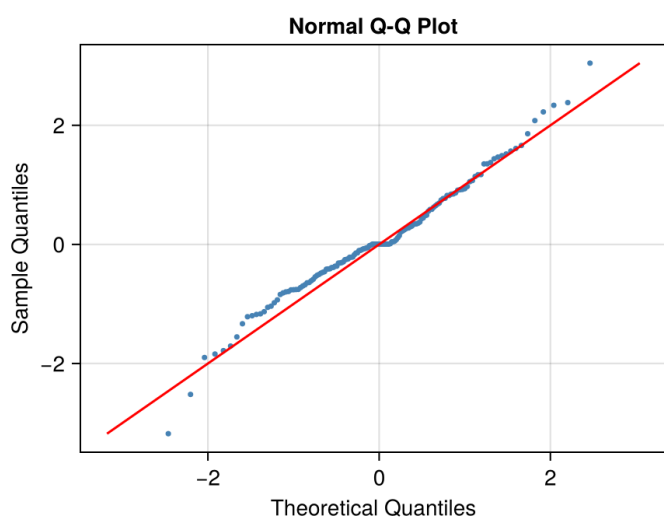
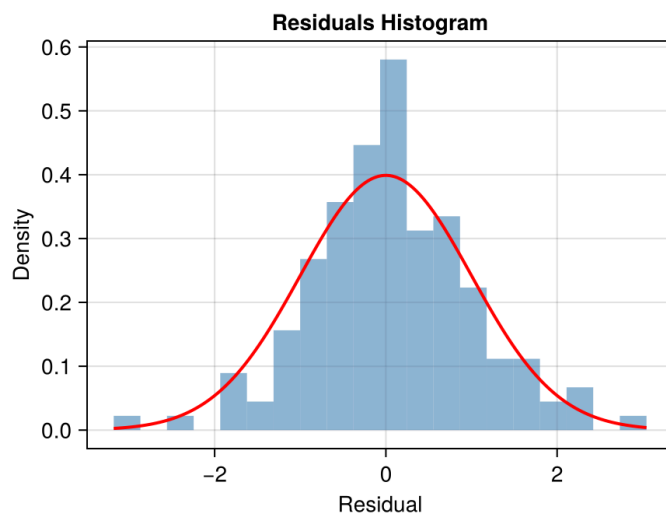
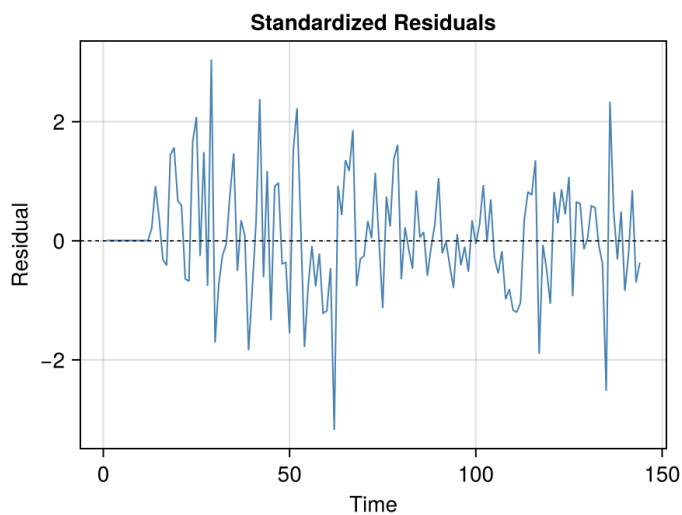
- All in all, these results are grossly compatible wrt to those suggested by the ACF/PACF plots.

```
1 kf = kalman_filter(modelar);
```

```

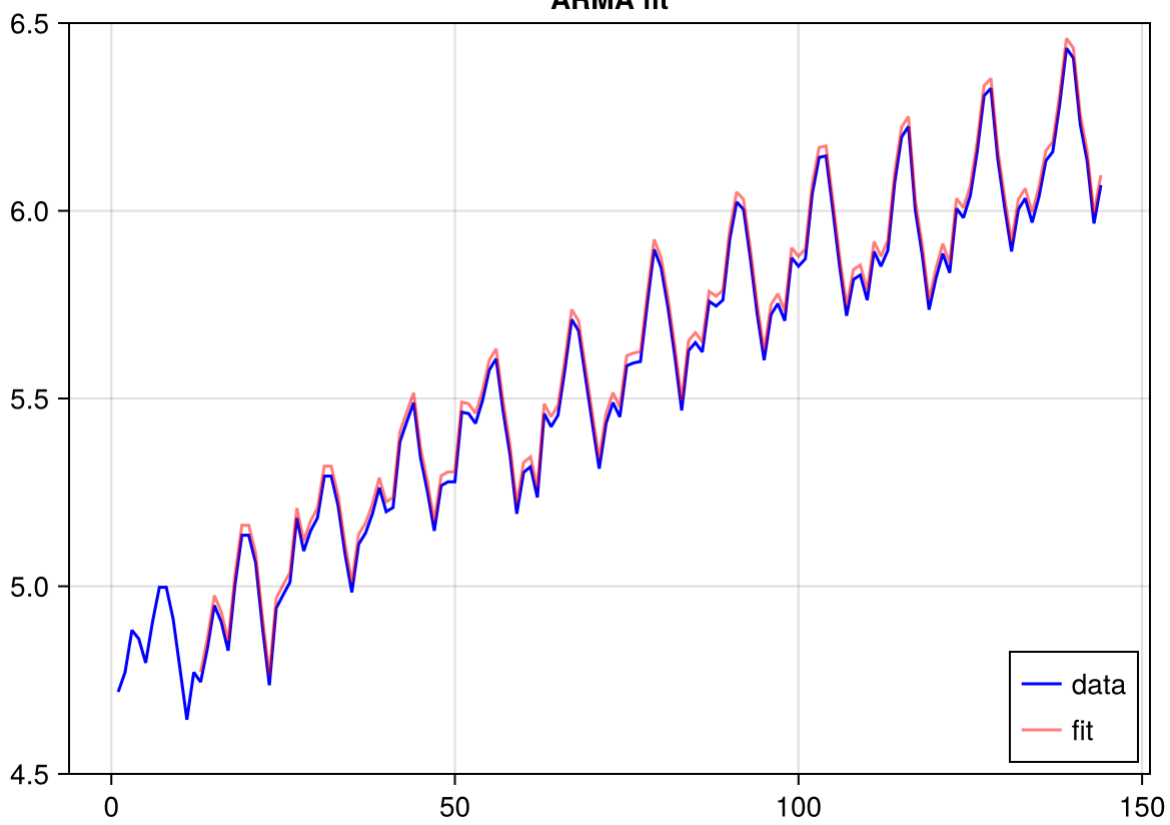
1 begin
2   # --- Estrai i residui standardizzati ---
3   residuals = [kf.v[t][1] / sqrt(kf.F[t][1,1]) for t in 1:length(kf.v)]
4   n = length(residuals)
5   tkf = 1:n
6
7   # --- Calcola ACF dei residui ---
8   max_lag = min(20, n ÷ 4)
9   acf_vals = autocor(residuals, 0:max_lag)
10  conf_bound = 1.96 / sqrt(n) # banda di confidenza 95%
11 end;

```



- The fit is not perfect, in particular for the tail values, yet for this exercise is fine.

ARMA fit



- Indeed the modeling of the input dataset is rather satisfactory.

Tip

Final remark!

State space models

- SSM form a broader approach to parametric modeling of complicated time series, where autoregressive and other stochastic behaviors can be combined with linear or nonlinear deterministic trends or periodic components.
- Functional behaviors can be linear or nonlinear, and can involve derivatives of state variables. Noise can be Gaussian or non-Gaussian, white or autoregressive.
- State space models are defined hierarchically with one level describing relationships between “state variables” defining the temporal behavior of the system and other levels describing how the underlying system relates to the observables.
- A state-space model for a (possibly multivariate) time series $\{Y_t, t = 1, 2, \dots\}$ consists of two equations. The first, known as the observation equation, expresses the w -dimensional observation Y_t as a linear function of a v -dimensional state variable X_t plus noise.

$$Y_t = G_t X_t + W_t, \quad t = 1, 2, \dots$$

- The second equation is called the state equation, and determines the state X_{t+1} at time $t + 1$ in terms of the previous state X_t and a noise term:

$$X_{t+1} = F_t X_t + V_t, \quad t = 1, 2, \dots$$

- The noise terms are often (but not necessarily) supposed to be uncorrelated.
- It is possible to find a state-space representation for a large number of time-series.
- Neither X_t nor Y_t is necessarily stationary.
- The beauty of a state-space representation lies in the simple structure of the state equation, which permits relatively simple analysis of the process X_t .
- The behavior of Y_t is then easy to determine from that of X_t using the observation equation.
- The coefficients of the model are calculated by least squares or maximum likelihood estimation, often through a recursive algorithm such as, e.g., the *Kalman filter*.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Ivezić et al. \(2020\) - "Statistics, Data Mining, and Machine Learning in Astronomy"](#)
- [Feigelson et al. \(2016\) - "Autoregressive Times Series Methods for Time Domain Astronomy"](#)
- [Feigelson & Babu \(2013\) - "Statistical Methods for Astronomy"](#)

Further Material

Papers for examining more closely some of the discussed topics.

- [Alexander \(1997\) - "Is AGN Variability Correlated with Other AGN Properties? ZDCF Analysis of Small Samples of Sparse Light Curves"](#)
- [Kelly et al. \(2014\) - "Flexible and Scalable Methods for Quantifying Stochastic Variability in the Era of Massive Time-domain Astronomical Data Sets"](#)
- [Stone et al. \(2022\) - "Optical variability of quasars with 20-yr photometric light curves"](#)
- [Tarnopolski et al. \(2020\) - "A Comprehensive Power Spectral Density Analysis of Astronomical Time Series. I. The Fermi-LAT Gamma-Ray Light Curves of Selected Blazars"](#).

Credits

This notebook contains material obtained from <https://www.analyticsvidhya.com/blog/2018/09/non-stationary-time-series-python/>, <https://towardsdatascience.com/how-to-analyse-a-single-time-series-variable-11dcca7bf16c>, <https://machinelearningmastery.com/grid-search-arma-hyperparameters-with-python/>, <https://machinelearningmastery.com/arma-for-time-series-forecasting-with-python/>, <https://www.analyticsvidhya.com/blog/2016/02/time-series-forecasting-codes-python/> and from <https://www.lem.sssup.it/phd/documents/Lesson6.pdf>.

Course Flow

.....

	Previous lecture	Next lecture	Course Summary
notebook	Science case about climate data	Science case about AGN and blazars	Course Summary
html	Science case about climate data	Science case about AGN and blazars	Course Summary

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Notebook v1.0.0 - 16 April 2026